Last Mile Delivery Capacity Planning with Two-Sided Uncertainty

Abstract

Crowdsourced delivery (CD) is increasingly combined with private delivery (PD) to respond to last mile delivery (LMD) demand uncertainty. As independent contractors, CD drivers self-schedule, which simultaneously creates labor supply uncertainty. Consequently, such LMD capacity planning requires addressing both uncertainties concurrently, a duality we term "two-sided uncertainty." In this setting, two key questions emerge: what is the optimal level of PD capacity when CD is available? How does two-sided uncertainty impact the PD capacity decision? To that end, we adapt a two-stage newsvendor model to explore the impact of two-sided uncertainty on optimal PD capacity when CD is available. We combine Monte Carlo simulation and optimization in empirically-grounded experiments using data from a Brazilian prepared meals company. We find a positive relationship between two-sided uncertainty and PD capacity. We also reveal nuance in the relationship by identifying moderating effects due to correlation in driver availability across CD price tiers and failed delivery costs. The results also indicate that optimal delivery capacity typically combines PD and CD sources rather than relying on a single source. We show that managerial decisions based solely on lowest unit delivery cost while assuming away two-sided uncertainty can lead to suboptimal decisions.

Keywords: Last mile delivery, uncertainty, capacity management, crowdsourced delivery, newsvendor, direct-to-consumer

1 INTRODUCTION

Last mile delivery (LMD) is an essential yet costly service in direct-to-consumer (DTC) business (Masorgo et al. 2024, Nguyen et al. 2019). Spurred by growth in online sales and amplified during the COVID-19 pandemic, the market for LMD service is expected to exceed \$200B by 2027 globally (Placek 2022). As DTC becomes a permanent feature of consumer-centric supply chains (Baldi et al. 2024, Esper et al. 2020, Wang et al. 2022), companies face increasing pressure to optimize LMD capacity to remain competitive (Tokar et al. 2020, Snoeck et al. 2020). However, LMD capacity planning is complicated by uncertainties in both demand and labor supply, making it difficult to effectively and efficiently serve customers (Winkenbach and Janjevic 2018, Cohen and Zhang 2022, Behrendt et al. 2024, Luy et al. 2024).

The challenges in LMD capacity planning are exemplified in the case of a Brazilian DTC company, hereafter referred to as *AffordableMeals*[1], which produces and delivers prepared meals directly to online customers. In a strategic shift in early 2024, the company added lower-priced products to its offerings to drive growth. This strategic shift resulted in substantial increases in both order volume and variability, transforming what had been a manageable DTC scheduling activity into a complex LMD capacity planning problem. *AffordableMeals* found their existing private delivery (PD) capacity insufficient to handle the increased (and volatile) demand, so they incorporated a low-cost crowdsourced delivery (CD) option as a source of flexible, on-demand LMD capacity. However, this option proved challenging due to unpredictability in the CD labor supply. As shown in this example, the *AffordableMeals* case illustrates how uncertainty in both demand and CD labor supply complicates LMD capacity planning (see Figure 1).

[———*Insert Figure 1 Here*———]

For *AffordableMeals*, LMD options include investing in either PD capacity, CD capacity, or adopting a hybrid approach that augments PD capacity with CD capacity (Garland 2022, Repko 2022, Straight 2022, Castillo et al. 2021, Luy et al. 2024, Behrendt et al. 2024). In the PD-only option, the company directly controls and operates a fixed delivery capacity. Generally, the PD option may consist of employed drivers or dedicated delivery capacity contracted through a third-party provider. PD is characterized by a financial commitment made to either the employees or the third-party provider. While a PD arrangement offers the benefits of known driver availability and cost structure, it constrains the ability of the company to adjust

[1] The company is anonymized for confidentiality purposes.

its LMD capacity in the short-run. Such inflexibility creates risk when delivery demand varies significantly (Suzuki et al. 2025). In the case of *AffordableMeals'* LMD capacity planning, if demand is underestimated, they may experience failed deliveries, thus incurring additional costs related to re-delivery and customer compensation, as well as the loss of consumer goodwill. Conversely, if demand is overestimated, they may incur expenses for idle PD drivers that are contracted for the week. Simply put, demand-capacity mismatches may result in either failed delivery costs or asset under-utilization, making delivery demand uncertainty an essential consideration in LMD capacity planning.

When a CD option is available, companies can add flexible LMD capacity that can reduce dependence on inflexible PD capacity (Qi et al. 2018, Castillo et al. 2018, Macrina et al. 2020, Akeb et al. 2018, Devari et al. 2017, Ta et al. 2023). CD drivers can be contracted on-demand and often at lower costs than PD capacity, although this may introduce operational challenges, as illustrated by the *AffordableMeals* case. After their strategic shift to lower-priced products, the company found that augmenting with CD capacity could help manage unpredictable demand. However, the reliability of CD service was questionable, as CD drivers had inconsistent availability or frequently rejected routes. Figure 2 depicts the difference in variability between PD and CD capacity, with CD capacity showing higher volatility than PD. These CD labor supply issues led the company to frequently make costly real-time adjustments, such as converting planned, cost-efficient car routes into on-demand, less cost-efficient motorcycle deliveries, a premium CD service. The *AffordableMeals* case shows how CD capacity, despite its potential for cost savings and flexibility, introduces new labor supply uncertainties that must be carefully balanced against stable PD capacity.

[———*Insert Figure 2 Here*———]

The logistics and supply chain management (LSCM) literature agrees that using CD drivers adds complexity to LMD capacity planning (Masorgo et al. 2024). As independent contractors, CD drivers self-schedule by choosing when to work, which CD platforms to partner with, and which LMD tasks to accept (Saunders et al. 2025, Le et al. 2019). CD drivers also vary in decision-making behavior, with some indiscriminately accepting all tasks and others only accepting tasks when surge pricing is implemented (Castillo et al. 2022). These issues elevate the impact of CD labor supply uncertainty, which manifests as increased risk of unfilled routes, delivery delays, and the need for real-time operational adjustments. For example, even though CD platforms may dynamically increase driver compensation during the day to boost CD labor supply, they may still face the risk of a labor shortage (Cachon et al. 2017, Gdowska et al. 2018, Huang and

Ardiansyah 2019). There is also risk of delivery delays due to human errors (Awaysheh et al. 2021), since the pool of potential CD drivers includes amateurs with little experience (Castillo et al. 2022). Additionally, CD platform competition for labor results in variability in driver pool size across platforms, as each may use a different remuneration strategy (Lee et al. 2022). Consequently, using CD creates a duality, where LMD capacity planning must consider CD labor supply uncertainty alongside delivery demand uncertainty. We term this duality "two-sided uncertainty."

DTC companies thus face a critical challenge of determining how many deliveries to assign to PD capacity when CD capacity is available. *AffordableMeals* regularly faces this challenge in their weekly capacity planning process, which relies on basic demand and labor supply averages and managerial judgment. This naïve approach has led to suboptimal decisions of either over- or under-allocating PD capacity. With an average of 760 deliveries per day spread across different delivery modes and CD price tiers[2], a more systematic approach to decide between PD capacity, CD capacity, or a hybrid model can improve efficiency. The LSCM literature suggests that classic inventory theory can address this LMD capacity planning problem. Specifically, viewing this managerial problem as balancing PD capacity to serve anticipated demand with CD capacity to react to unanticipated demand to minimize total delivery cost, we can model the problem as a two-stage newsvendor problem (NVP2S), originally formulated and solved by Chung and Flynn (2001). The NVP2S model allocates primary capacity based on historical demand distributions in the first stage. Then, it uses secondary capacity to meet additional demand that arises in the second stage. Applied to *AffordableMeals'* context, the key managerial decision is determining how much PD capacity to secure, with CD being available to meet remaining demand (see Figure 1).

However, despite the promising approach offered by the NVP2S framework, it includes restrictive assumptions that do not necessarily reflect the reality of DTC companies. The NVP2S assumes secondary capacity is both deterministic and more expensive than the primary capacity. However, the *AffordableMeals* case shows that CD capacity has highly uncertain labor availability and CD delivery cost may fluctuate, resulting in either lower or higher costs relative to PD costs depending on market conditions. Therefore, we adapt the NVP2S by relaxing its restrictive assumptions to account for two-sided uncertainty. Our adaptation of the NVP2S model incorporates both a stochastic CD labor supply and the possibility that CD capacity may cost less than PD capacity. These modifications better reflect the real-world complexity of

[2] We use the term "CD price tier" or simply "tier" to connote that CD capacity can be acquired from different modes within a platform, such as standard car or premium motorcycle delivery, at different prices, like in the case of *AffordableMeals*.

DTC operations where CD capacity is neither consistently available nor always more expensive.

We assess the effect of two-sided uncertainty on optimal PD capacity by applying the adapted NVP2S model in a series of experiments that combine Monte Carlo simulation and optimization techniques populated with LMD data from *AffordableMeals*. The analysis illustrates our adapted model's usage in practice and reveals insights into the context-dependent impact of two-sided uncertainty on optimal PD capacity in LMD planning. We vary demand and labor supply uncertainty while incorporating important contextual variables to understand their impact on the model's output: the number of deliveries to be made using PD capacity. Specifically, we include 1) the degree of correlation in the availability of drivers across different tiers of CD labor (e.g., during adverse weather), and 2) cost penalties for failed deliveries.

Our study contributes a novel perspective on LMD to the LSCM literature by inductively developing theory from quantitative observations of *AffordableMeals'* LMD services (Stank et al. 2024). Through analyzing empirical data and adapting a classic inventory theory model to match real-world complexities, we propose several contextualized relationships between two-sided uncertainty and optimal PD capacity. First, we found that while increased two-sided uncertainty generally requires greater PD capacity allocation, this relationship becomes nuanced at different demand and CD labor supply uncertainty levels. Specifically, our analysis revealed a U-shaped relationship between demand uncertainty and optimal PD capacity when CD labor supply uncertainty is high. More PD capacity is needed at both low and high demand uncertainty levels because the high CD capacity uncertainty raises the risk of failed deliveries, despite demand predictability. However, less PD capacity is needed at moderate demand uncertainty levels since the cost of idle PD capacity is more expensive than the cost of failed deliveries, making it more economical to rely on CD despite the high uncertainty.

Second, the analysis also revealed how correlation in CD labor supply availability across CD price tiers impacts LMD capacity planning. When CD labor supply shows high positive correlation (e.g., when weather affects all CD tiers similarly), companies can actually reduce their PD capacity since aggregate CD labor supply becomes more predictable. However, this relationship reverses when failed delivery costs are high, as more PD capacity is needed to buffer against simultaneous CD labor shortages that would lead to failed deliveries.

Finally, our results consistently showed that hybrid delivery models combining PD and CD capacities outperformed single-source approaches in minimizing total expected LMD cost. This finding emerged directly from *AffordableMeals'* LMD data. From these empirical observations, we develop five testable propositions about two-sided uncertainty and PD capacity optimization when CD is available. A six month follow-up with *AffordableMeals* after their strategic shift to lower-priced products revealed that they had independently evolved toward a hybrid strategy through trial-and-error experimentation. This independently realized outcome aligns with our findings, thus providing external validation to our research.

2 LITERATURE REVIEW AND THEORETICAL FOUNDATIONS

2.1 Crowdsourced Delivery

Our research contributes to the growing literature on crowdsourced delivery (CD) in last mile logistics. We examine how two-sided uncertainty emerges from both demand and CD labor supply variability, how uncertainty has been addressed in the literature, and the consequences for LMD capacity planning decisions. This dual uncertainty has become increasingly relevant as companies adopt hybrid delivery models that combine PD and CD capacity (Behrendt et al. 2024, Luy et al. 2024).

The analytical CD literature has predominantly focused on optimizing LMD that incorporates CD through various modeling approaches. For instance, Yang et al. (2024), Arslan et al. (2019), Archetti et al. (2021) and Macrina et al. (2020) propose variations of the seminal vehicle routing problem (VRP) by Dantzig and Ramser (1959) to incorporate the use of CD drivers. The treatment of uncertainty varies across these studies: some assume both demand and CD labor supply are deterministic for mathematical tractability (e.g., Macrina et al. 2020), while others model demand as stochastic but CD labor as deterministic (e.g., Yang et al. 2024, Archetti et al. 2021), and still others capture both stochastic demand and CD labor supply (e.g., Arslan et al. 2019, Gdowska et al. 2018). Similar studies developing scheduling algorithms for matching CD drivers with delivery tasks (e.g., Meng et al. 2024, Chen et al. 2016, Zhang et al. 2019) or optimizing pricing with CD (e.g., Dayarian and Pazour 2022, Fatehi and Wagner 2021) have also incorporated stochasticity in either demand or CD labor supply.

The empirical CD literature has examined the strategic and operational merits of CD, developing guidance for how to implement CD in practice (e.g. Rougès and Montreuil 2014, Simoni et al. 2020, Buldeo Rai et al. 2017). The advantages of CD that appeal to DTC companies include increased flexibility in last mile logistics (Devari et al. 2017), lower fixed costs due to reductions in asset ownership (Macrina et al. 2020, Akeb et al. 2018), and improved service quality perceptions (Ta et al. 2023). The CD literature has also revealed the challenges in obtaining such advantages. For instance, studies examining customer perceptions of CD reveal that the service quality depends on how platform technology design facilitates customer-driver interactions (Ta et al. 2018, Merkert et al. 2022, Wang et al. 2024, Ta et al. 2025). Indeed, CD driver behavior impacts customer service perceptions (Ta et al. 2025), with negative attitudes and inflexibility harming the customer experience (Masorgo et al. 2023). This stream of research highlights the operational benefits and challenges that arise from relying on independent CD drivers.

The empirical literature also agrees that CD driver self-scheduling behavior fundamentally alters labor supply dynamics in LMD. As independent contractors, CD drivers have autonomy in choosing when and how long to work, and which deliveries to accept (Carbone et al. 2017, Castillo et al. 2018, Ta et al. 2018, Fatehi and Wagner 2021, Benjaafar and Hu 2020, Le et al. 2019, Xu et al. 2023). This autonomy induces CD labor supply uncertainty, which manifests as varying task acceptance rates (Xu et al. 2023, Castillo et al. 2022) and reliability issues (Gdowska et al. 2018, Huang and Ardiansyah 2019). CD platforms often attempt to manage this uncertainty by dynamically adjusting driver pay to match labor supply with demand (Hu et al. 2021, Garg and Nazerzadeh 2022, Guda and Subramanian 2019, Besbes et al. 2021, Castillo et al. 2024). However, even with dynamic pricing, labor supply is still variable as drivers respond differently to incentives and platform competition affects available driver pools (Lee et al. 2022).

Customer-focused research on CD reveals how demand uncertainty compounds these labor supply challenges. Studies examining customer attitudes and behaviors show that delivery demand varies significantly by time of day, weather conditions, market dynamics (Merkert et al. 2022), and service quality outcomes (Masorgo et al. 2024, Baldi et al. 2024). This demand variability creates particular challenges for DTC companies, as CD platforms must simultaneously manage fluctuating customer demand and uncertainty in driver availability (Luy et al. 2024, Behrendt et al. 2024, Dayarian and Savelsbergh 2020). The complexity amplifies in urban areas where multiple delivery platforms compete for both customers and drivers (Rose et al. 2023, Castillo et al. 2022), creating additional variability in both demand and labor supply.

Some studies have explored or proposed hybrid delivery models that combine PD and CD capacity (e.g., Kafle et al. 2017, Qi et al. 2018, Castillo et al. 2021, Luy et al. 2024, Behrendt et al. 2024). While these studies provide valuable insights into hybrid delivery operations, there remains a critical gap in understanding how demand and CD labor supply uncertainty interact to affect LMD capacity planning decisions. As companies increasingly adopt hybrid delivery models, understanding this two-sided uncertainty becomes crucial for effective and efficient LMD.

The newsvendor problem (NVP) is a classic inventory theoretical perspective, which offers a promising framework for addressing this gap, as it can guide LMD capacity decisions under uncertainty. Chung and Flynn's (2001) two-stage newsvendor problem (NVP2S) can guide thinking about LMD capacity planning. However, it assumes deterministic secondary capacity and higher secondary source costs. These assumptions must be relaxed to accommodate two-sided uncertainty in LMD, since CD is neither consistently available nor always more expensive than PD capacity. By adapting the NVP2S to account for these real-world considerations, we can better understand how companies can balance PD and CD capacity under two-sided uncertainty.

2.2 The Newsvendor Problem

The NVP provides the theoretical foundation for our research. Since its introduction by Arrow et al. (1951), the NVP has been foundational in guiding production and capacity decisions under uncertainty in various settings. In the original NVP, a decision-maker must decide on an order quantity for a single product sold in a single season, given that demand is uncertain but follows a known distribution. The objective of the NVP is typically to minimize cost (or maximize profit), where a critical fractile function prescribes a production quantity based on production and capacity costs along with demand and supply parameters. Extensions of the NVP include multiple ordering periods (Karmarkar 1987, Bisi et al. 2011), multiple products (Zhang 2010, Zhang and Du 2010), environmental sustainability (Raz et al. 2013), profit risk constraints (Lin et al. 2022), and, importantly for our problem, supply uncertainty (Dada et al. 2007, Burke et al. 2009, Käki et al. 2015). Comprehensive reviews of the NVP literature are provided by Qin et al. (2011), Choi (2012), Khouja (1999) and Zhang and Siemsen (2019).

2.2.1 Multi-Period Newsvendor Problems.

The multi-period NVP is particularly relevant to our research since it allows for capacity adjustments as demand uncertainty is observed across multiple periods. Based on the demand distribution and problem parameters, there is an initial period when a basic amount of products or services are created. In subsequent periods, additional amounts of products or services are created as actual demand becomes known. Chung and Flynn's (2001) two-stage newsvendor problem (NVP2S) addresses single product production and balancing lower-cost anticipatory production capacity with higher-cost reactive capacity. Extensions of the NVP2S have considered multiple products (Reimann 2011), capacity setting rather than order quantities (Cattani et al. 2008), and different supplier characteristics (Allon and Van Mieghem 2010, Janakiraman et al.

2015). However, these models maintain the assumption that secondary capacity is both deterministic and more-expensive, which does not hold in CD contexts.

2.2.2 Supply Uncertainty in the NVP.

Supply uncertainty in the NVP is typically modeled as a level of availability determined by supplier reliability. Dada et al. (2007) consider the NVP with multiple suppliers, where each supplier is either unreliable or perfectly reliable. The authors show that lower cost suppliers are preferred over higher cost suppliers regardless of their reliability. Burke et al. (2009) consider a choice between single and multiple suppliers in an environment of demand and supply uncertainties and investigate the allocation of orders among multiple suppliers. Käki et al. (2015) provide a closed form solution for stochastic, interdependent demand and supply, however, both the demand and supply are distributed uniformly. They suggest that the impact of supply uncertainty has a significant impact on newsvendor decisions, especially under a high-profit condition. Ma et al. (2016) consider both supply and demand uncertainty when the supplier is only paid for the quantity that is actually delivered. However, these models typically assume a small number of suppliers with known reliability levels. This contrasts with CD, where numerous independent drivers self-schedule. Our research extends the NVP2S to accommodate the realities of two-sided uncertainty last-mile delivery.

3 METHODOLOGY: LAST MILE DELIVERY CAPACITY PLANNING UNDER TWO-SIDED UNCERTAINTY

3.1 Capacity Planning Problem Description and Formulation

We examine hybrid LMD capacity planning from the perspective of *AffordableMeals*, which oversees an average of 760 daily deliveries. The logistics managers determine PD capacity on a weekly basis using basic descriptive statistics about LMD demand, CD labor supply, and judgment-based heuristics about CD availability. Let the number of LMDs to be made each day be represented by a random variable D (the notations used herein are presented in Table 1). Since LMD demand naturally follows patterns of normal variation around a central tendency, we model it as normally distributed ($D \sim Norm(\mu, \sigma)$).

A LMD is made by either a PD driver or a CD driver. The company incurs a labor cost for each PD driver employed and an operating cost component while making deliveries. Let l represent the labor cost, which includes PD driver wage and benefits in the case of employed drivers or a fixed fee in the case of dedicated capacity provided by a third-party. This cost l is incurred regardless of whether deliveries are made. In turn, let v represent operating costs, which are incurred only if the PD driver makes a delivery. The costs v consist of maintenance, fuel, and amortization costs. Both cost parameters l and v are accounted for on a per-delivery basis. Let a represent the per-delivery cost using PD capacity. This cost is defined as $a = l + v$. PD capacity drivers accept all delivery tasks assigned to them, although not all tasks may be satisfactorily completed. For instance, traffic conditions may result in deliveries taking longer than planned, or customers may not be home to accept a delivery when such a condition is required. Therefore, the reliability of PD drivers is an important problem feature to account for. Let r_{priv} represent the probability of a PD capacity driver completing a delivery task assigned to them, where $0 < r_{priv} \leq 1$.

[———*Insert Table 1 Here*———]

CD drivers, who work as independent contractors, differ from PD capacity drivers in their decision-making. CD drivers' willingness to accept delivery tasks can vary widely, with some being highly selective and requiring higher pay and others indiscriminately accepting delivery tasks for lower pay (Castillo et al. 2022). Consequently, the CD labor supply — measured as the number of deliveries available for completion — can vary and depend heavily on offered remuneration. Let n represent the number of price tiers (or sources) of CD labor supply, each with a corresponding pay amount defined as b_i. The CD driver remuneration structure can have fixed and variable components (Huang and Ardiansyah 2019) or just a variable component (Castillo et al. 2018, Guo et al. 2019). Let $b_{f,i}$ represent a fixed component of CD compensation, and let $b_{v,i}$ represent a variable component of CD compensation. Therefore, $b_i = b_{f,i} + b_{v,i}$.

Each price tier of CD labor supply $i \in N = \{1, 2, \ldots, n\}$ has a delivery capacity measured as number of deliveries that can be made for the planning period for that price. Let B_i represent the CD capacity in price tier $i \in N = \{1, 2, \ldots, n\}$. CD drivers willing to accept the remuneration amount b_i will make the deliveries in tier B_i. The CD capacity available at each B_i for cost b_i is uncertain and likely to fluctuate daily. Therefore, we also treat B_i as a random variable to account for uncertainty in CD labor supply. In practice, CD capacity is bounded by the finite pool of available drivers in a given market on any given day, we model it using Beta distributions ($B_i \sim Beta(\alpha_i, \beta_i)$), which provide a flexible bounded continuous distribution appropriate for representing capacity constraints.

External factors, like social events and adverse weather events, may simultaneously impact CD drivers' availability across all platforms and price tiers. In the case of *AffordableMeals*, management discovered systematic CD availability patterns through their weekly planning process. For instance, CD reliability drops significantly on Saturday evenings when social events reduce driver availability across all platforms. Weather conditions also create correlated effects: during rain or other adverse conditions, the company sees reduced CD availability across CD price tiers. These examples illustrate how CD labor supply B_i across each price tier b_i may rise or fall collectively. To describe the relationship between CD labor supply across tiers, let ρ represent CD labor supply correlation, which captures the fact that CD labor supply is interconnected across B_i.

A distinctive feature of LMD capacity planning when CD is available is the labor supply function. Following neoclassical theory, higher compensation is typically assumed to lead to greater labor supply. This assumption forms the basis for dynamic or "surge" pricing strategies, where drivers are paid higher compensation during periods of heightened or urgent demand to increase CD labor (Castillo et al. 2024, Miao et al. 2023, He et al. 2022, Hu et al. 2021, Besbes et al. 2021). However, the population of CD drivers is heterogeneous regarding compensation acceptability, with some drivers accepting any offer and others only accepting delivery tasks when prices are raised (Castillo et al. 2022). Hence, a labor supply continuum exists, where some CD drivers will accept delivery tasks at any payment and others will only accept delivery tasks when payments increase substantially.

Similar to dynamic pricing, *AffordableMeals'* CD labor supply function reflects the acquisition of different modes of CD at increasing price tiers from a single CD platform. When *AffordableMeals* struggles to acquire sufficient CD capacity at lower price tiers, they incrementally move to the next higher-priced CD tier until a CD driver accepts the task. To model this, we arrange CD tiers by price, such that $b_i < b_{i+1} < \cdots < b_n$. Further, we assume that higher tiers allow access to new set of drivers willing to accept a delivery at b_i, leading to a CD labor supply curve as depicted in Figure 3. Each price b_i thus corresponds to a specific CD capacity, B_i, available from that tier. When CD capacity at tier B_i is exceeded and insufficient to meet demand D, CD task remuneration increases to b_{i+1} to access the next available CD capacity amount B_{i+1}. While the per-delivery task remuneration amounts b_i, and b_{i+1} are universally available to all CD drivers, only the subset of drivers willing to accept tasks at b_i and b_{i+1} contribute CD capacity at B_i and B_{i+1}, respectively. This approach also accommodates a variety of CD remuneration structures, in which drivers are compensated on a fixed piece-rate, time- or mileage-variable rate, or a combination. This approach also allows for reference-dependent behavior which may not be a monotonically increasing function (Miao et al.

2023).

[———*Insert Figure 3 Here*———]

CD drivers' freedom to reject delivery task offers is also a critical problem feature. A delivery task may not be made on time because a CD driver rejected the offer or because no CD driver was available to accept it. Furthermore, since entry barriers to becoming a CD driver are relatively low, not all workers have the professional experience of full-time PD drivers and may make errors resulting in failed delivery tasks. Thus, each CD driver may not be 100% reliable. Let r_{CD} represent the reliability of CD drivers in completing a delivery task, where the probability that a CD task is completed satisfactorily is $0 < r_{CD} \leq 1$. For example, in *AffordableMeal's* operations, CD drivers may not have consistent availability throughout the day or may reject routes due to distance, timing, or compensation, requiring real-time adjustments such as converting car routes to multiple motorcycle routes, which may lead to failed deliveries. The company incurs additional costs associated with a failed delivery such as re-delivery costs, returning the package to the warehouse, additional storage, and customer compensation. Let u represent the cost penalty for a failed delivery.

Based on the probability distributions of delivery demand, D, and available capacity across CD tiers B_i, acquired at cost b_i, the goal of the model is to decide how many deliveries to allocate to PD capacity, with the balance being fulfilled through CD. Let Q^* represent the cost-minimizing PD capacity, measured as the number of deliveries, with the corresponding unit delivery cost, a, measured on a per-delivery basis. This also allows for an equivalent comparison between the PD and CD capacity. The allocation decision is made in two stages. First, the PD capacity, Q^*, is selected in the first stage using historical data. In the second stage, a demand of D and the available CD capacity B_i of tier $i = 1, 2, \ldots, n$ is observed. Received deliveries are then allocated to PD capacity, with the balance being offered to available CD drivers.

3.2 Adapting the NVP2S for Two-Sided Uncertainty

LMD capacity planning considers both PD and CD sources of capacity, as illustrated by *AffordableMeals'* use of fixed PD capacity per period, supplemented by flexible CD capacity. The incorporation of CD into LMD requires a model that accounts for dual sourcing. As such, we adapt Chung and Flynn's (2001) NVP2S model, which originally addressed how to balance cheap anticipatory production with expensive reactive production. Our LMD capacity planning problem is consistent with their original formulation, with PD capacity representing anticipatory capacity and CD capacity representing reactive capacity.

Two assumptions in the NVP2S must be relaxed to accommodate two-sided uncertainty. First, the original NVP2S model assumed guaranteed availability of reactive capacity, which is not a realistic assumption with uncertain CD labor supply. CD drivers' self-scheduling behavior means their availability is not stable and often correlated across different price tiers (e.g., when bad weather affects all drivers simultaneously). Second, unlike the original NVP2S model which assumed reactive capacity always costs more than primary capacity, some CD options (e.g., low-cost motorcycle delivery) may cost less than PD capacity. *AffordableMeals* regularly uses these cheaper CD tiers for smaller deliveries, requiring a modification to how the model allocates demand between PD and CD.

The cost structure in our adapted model reflects the actual components *AffordableMeals* accounts for. PD capacity incurs both labor costs (which must be paid even when drivers are idle) and operating costs (fuel, maintenance, etc.). CD capacity costs vary by delivery mode, with motorcycles typically being cheapest, followed by cars, vans, and premium (speedy) motorcycles. Failed deliveries create additional costs through product returns, redelivery attempts, and customer compensation. The company's experience showed these costs are particularly significant for temperature-controlled deliveries where product spoilage is a concern.

Our adaptation of the NVP2S model optimizes the LMD capacity at the system level by selecting the optimal PD capacity that balances the various costs with two-sided uncertainty. The model first identifies which CD tiers cost less than PD capacity and allows them to be used first. It then optimizes how much demand to allocate to PD capacity versus the more expensive remaining CD options. The model uses numerous probability distributions to account for demand uncertainty and CD labor supply uncertainty in each CD tier.

The mathematical complexity introduced by these real-world conditions precludes derivation of a closed-form solution, as it involves nested convolutions of random distributions alongside potential correlations between CD labor supply tiers. Therefore, we implement the model through simulation optimization (Carson and Maria 1997, April et al. 2003, Amaran et al. 2015), which combines Monte Carlo simulation with the adapted NVP2S optimization framework to identify optimal PD capacity allocation decisions. This approach systematically generates random samples from probability distributions to simulate demand and CD labor supply, which are then used to determine the PD capacity level that minimizes expected total delivery costs. The complete mathematical formulation and technical details of the Monte Carlo simulation optimization are provided in Appendix A.

3.3 Model Application and Monte Carlo Simulation Optimization Experiments

The preceding sections define the problem and our adaptation of the original NVP2S model to a LMD capacity planning context. To demonstrate our model's practical application and develop theoretical insights about the role of two-sided uncertainty in LMD capacity planning, we conduct a series of Monte Carlo simulation optimization experiments with empirical data from *AffordableMeals*. This research effort illustrates the applicability and versatility of our adapted model. The results of these experiments reveal how two-sided uncertainty affects the total expected cost of LMD operations.

3.3.1 Research Setting: Brazilian Prepared Meals Company

AffordableMeals operates in twelve Brazilian markets, offering delivery of frozen and fresh foods, snacks, and desserts to businesses and consumers. Their vertically integrated operation begins at a central production facility where meals are prepared and then undergo quick freezing to ensure freshness without preservatives. Third-party logistics providers transport finished products to distribution centers in each market. While these facilities operate as traditional distribution centers, they function more like dark stores, managing local inventory and order preparation. Orders placed through the company's e-commerce platform are then prepared for delivery throughout the service area.

Our analysis focuses on LMD in the company's largest market during March-April 2024, where the company manages an average of 759.9 daily deliveries. This single-market approach captures particular demand patterns, CD labor availability, and cost structures unique to one operating environment. Although neighborhood-level variations exist within this market, the company's planning process involves market-level capacity planning decisions. Importantly, the model developed in Appendix A is generalizable across markets, allowing *AffordableMeals* to implement it in each of the cities in which they operate. This approach balances contextual specificity with broad applicability to the company's full operations.

AffordableMeals allocates order delivery to one of two modes: PD drivers or CD drivers. The LMD capacity planning process relies primarily on heuristics and managerial judgment to make allocation decisions. Generally, CD had been favored since it has lower unit delivery costs than PD. However, due to lower reliability, planners consider failed CD deliveries from the previous week as they determine how much PD capacity to contract for in the coming week.

The PD mode is comprised of dedicated couriers that are contracted from a third-party provider, resulting in known, fixed capacity that makes PD the first choice in LMD planning. PD capacity is sourced on a weekly planning horizon, but is not required to be used. However, the company incurs a cancellation fee if it decides not to use some part of the already contracted PD capacity. This is akin to a scenario where a company employs drivers and incurs labor costs even when it decides to resort to a cheaper CD option.

AffordableMeals sources CD capacity from five different price tiers within a single CD platform. Thus, CD capacity is acquired on-demand at prices that depend on the service level and vehicle type, where drivers may operate motorcycles, cars, or vans to make deliveries. The advertised rate that *AffordableMeals* pays is structured as a combination of piece rate (dependent on number of stops) and distance (per-km) remuneration. *AffordableMeals* can source from five tiers of CD capacity with increasingly higher rates: low cost motorcycle couriers, medium cost cars or vans, premium rate motorcycle couriers, or super premium rate motorcycle couriers. Motorcycle and car delivery tends to be used the most despite vans having the potential for greatest route efficiency. This is because in meal delivery settings, vans offer slower deliveries in a dense urban environment. Further, a van that is loaded to capacity will result in later stops having lower quality meals, so vans are used sparingly. Based on historical driver payments maintained by *AffordableMeals*, an average per unit delivery cost can be calculated for each price tier[3] as shown in Table 2.

[———*Insert Table 2 Here*———]

A delivery can be unsuccessful if the customer rejects it or if the driver fails to reach the customer within the promised delivery window. The cost of a failed delivery adds to the total delivery costs due to the need to handle the returned order, reintegrate it into inventory, which is critical for temperature-controlled items, and attempt redelivery. The company estimates that the cost of a failed delivery is twice the average delivery cost.

The dataset provided by the company contains detailed delivery records including the delivery date, time, carrier, mode (i.e., motorcycle, car, or van), delivery cost per order, PD capacity, CD capacity by provider (e.g., motorcycle CD drivers, car CD drivers, and van CD drivers), delivery failure counts for PD and CD modes, and failed delivery costs. We supplemented the company's data with delivery expense data in Brazil, specifically, fuel, insurance, maintenance, and depreciation, to estimate local LMD operating costs (Callil et al. 2023). All values are reported in detail in Section 3.3.3.

[3] All cost values are reported in Brazilian Reals (R$).

3.3.2 Experimental Design

Table 3 outlines the Monte Carlo simulation optimization process. The model begins with generating daily delivery demand, D, which we base on empirical data provided by *AffordableMeals'* operations across three delivery windows (9am-1pm, 2pm-6pm, and 6pm-10pm). The next step is to create a CD labor supply, $B_i = (B_1, B_2, \ldots, B_5)$, measured as number of deliveries. The cost to access each tier of CD capacity is b_i, where $b_i < b_{i+1}$. As described in Section 3, we model the CD labor supply as tiered, whereby accessing additional tiers of CD labor can only be made if higher prices are paid. This accommodates the heterogeneous decision-making behavior of CD drivers (Castillo et al. 2022). Note that this modeling approach can also accommodate differing remuneration structures (e.g, fixed piece-rates, variable time or mileage rate, or combined structures). Each price tier b_i of CD labor supply has a mean capacity B_i, measured as the number of deliveries available.

[———*Insert Table 3 Here*———]

The model uses multiple empirical distributions of variables and constants to determine Q^*, the number of LMDs to allocate to the PD capacity. The model minimizes $H(Q)$ such that $H(Q) \geq \frac{u-a}{u-v}$. Q^* is the smallest Q that satisfies the critical fractile formula of the NVP2S (Chung and Flynn 2001, p. 756, eq. 4.4). The model uses a binary search to find Q^*. Upon determining Q^*, the PD cost can be calculated along with the CD delivery costs, yielding a total cost, $G(Q,D,B)$. The cost result of the iteration is recorded and then another model iteration is run. Upon completion of the runs, the results are compiled into a single dataset and analyzed using full factorial ANOVAs for each experimental scenario. The Monte Carlo simulation optimization was written in Java and run in AnyLogic 8.8.4. with the statistical analysis being conducted in R 4.3.3.

3.3.3 Experimental Variables

Our experiments examine the impact of the factors of interest on Q^* (see Table 4). The factors common to all experiments, *Demand Uncertainty* and *CD Supply Uncertainty*, are denoted by the coefficients of variation CV_D and CV_S, respectively. To operationalize CV_D, we change the standard deviation of D to achieve the range listed in Table 4. From the empirical data (see Table 5 for a summary of the empirical grounding), mean demand is $\mu = 759.9$ deliveries per day. We consider a standard deviation σ =

$\{0.0, 152.0, 304.0, 456.0, 607.9\}$ to set the levels of CV_D. To measure CV_S, we change the standard deviation of B_i to achieve the set of CD Labor Supply Uncertainty parameters listed in Table 4. From the empirical data, the mean CD capacity for each tier is $B_i = \{16.5, 389.8, 9.0, 67.5, 4.2\}$ deliveries. The derivation of the parameters used to achieve the CV_S values listed in Table 4 is explained in detail in Appendix B.

[———*Insert Table 4 Here*———]

[———*Insert Table 5 Here*———]

The Base Case scenario examines the effect of the relationship between two-sided uncertainty, $CV_D \times CV_S$, and optimal PD capacity, Q^*. In turn, Experiment 1 introduces as a factor *CD Supply Correlation*, ρ, which is the correlation in CD capacity across price tiers b_i (see Appendix B for a detailed explanation of how ρ is operationalized). This is an important experimental factor because dynamic market conditions or occurrences such as adverse weather or social events are likely to increase or decrease the availability of CD drivers across tiers simultaneously. Finally, Experiment 2 introduces *Failed Delivery Cost*, $u = \{\text{R\$}28.7, \text{R\$}35.1, \text{R\$}41.6\}$ as a factor. Both Base Case and Experiment 1 use the failed delivery cost provided by the company, $u = \text{R\$}35.1$. This approach is consistent with related studies (Guo et al. 2019, Archetti et al. 2021).

3.3.4 Experimental Constants

The constants used in the experiments are summarized in Table 5. We incorporate price tiers into the model as the cost b_i of acquiring a delivery at CD capacity tier $i \in N$. The values used are the per-delivery payment amounts offered by the platform: $b_i = \{\text{R\$}8.8, \text{R\$}10.9, \text{R\$}17.3, \text{R\$}22.0, \text{R\$}28.7\}$. Since the company does not include a fixed component for CD cost, $b_{f,i} = 0$. There are a total of $n = 5$ tiers of CD labor supply, which emulates having multiple tiers of CD drivers who only accept deliveries at increasingly higher prices, in accordance with Castillo et al. (2022). The model can incorporate more or fewer tiers, depending on the empirical context. Using the empirical data, the salvage value is calculated as $v = \text{R\$}17.6$. In all experiments, $r_{priv} = 0.986$, which is the average reliability for PD drivers.

3.4 Results

Each cell in the experimental design across all three experiments is replicated for 1000 iterations. Full factorial ANOVAs are conducted on each experiment's results to assess statistically significant differences in levels of the experimental variables described in Section 3.3.3. The results for the experiments are depicted in Figures 4 – 6.

3.4.1 Base Case: Two-Sided Uncertainty

In the Base Case, we examine Q^* as a function of Demand Uncertainty (CV_D) and CD Supply Uncertainty (CV_S). As shown in Figure 4, Demand Uncertainty accounts for a significant amount of variance in Q^* ($F(4) = 126.36, p < 0.01$). CD Supply Uncertainty (CV_S) also explains a statistically significant portion of the variance in Q^* ($F(4) = 4.89, p < 0.01$). However, the impact of CD Supply Uncertainty on Q^* is contingent upon the level of Demand Uncertainty, as shown by the significant $CV_D \times CV_S$ interaction ($F(16) = 4.14, p < 0.01$). This interaction is particularly evident at higher levels of CD Supply Uncertainty, as illustrated in Figure 4.

An interesting relationship is observed when CV_S is high. A U-shaped association between CV_D and Q^* emerges when $CV_S = 0.8$. At very high levels of CV_S the model recommends more PD capacity at low and high levels of CV_D. When demand uncertainty is low, the model recommends more PD capacity as a buffer against volatility in CD Labor Supply. When demand uncertainty is high, the risk of unmet demand is high so the model recommends more PD capacity to maintain service levels. When CV_D is moderate, the model favors CD capacity despite its high uncertainty, thus indicating that risk of idle, expensive PD capacity is greater than the risk of volatile CD labor supply.

[———*Insert Figure 4 Here*———]

3.4.2 Experiment 1: CD Labor Supply Correlation

In Experiment 1, we include CD Supply Correlation, ρ, as an experimental factor (refer to Figure 5). The variance in Q^* is significantly explained by Demand Uncertainty, CV_D ($F(4) = 269.70, p < 0.01$), and CD Supply Uncertainty, CV_S ($F(4) = 1380.97, p < 0.01$), as well as CD Supply Correlation ρ ($F(2) = 22469.28, p < 0.01$). The significance extends to all two-way interactions, including the interaction between Demand and CD Supply Uncertainties, $CV_D \times CV_S$ ($F(16) = 4.72, p < 0.01$), the interaction between Demand Uncertainty and CD Supply Correlation, $CV_D \times \rho$ ($F(8) = 23.21, p < 0.01$), and the interaction between CD Supply Uncertainty and CD Supply Correlation, $CV_S \times \rho$ ($F(8) = 409.57, p < 0.01$). Additionally, the three-way interaction among these factors is also significant ($F(32) = 5.04, p < 0.01$), indicating that the impact of two-sided uncertainty on Q^* varies with the degree of correlation in CD labor supply.

Two general trends are observed in Fig. 5: as CV_D and CV_S increase, so too does Q^*, indicating that more PD capacity is needed as a buffer against increases in demand and CD labor supply uncertainty. As the correlation in CD labor supply increases from 0.0 to 0.5, variability across n decreases. In effect, the CD labor supply is more synchronized across tiers, leading to improved predictability and stability of the CD workforce. As a result, the model prescribes greater reliance on CD over PD capacity. However, when CV_S increases to 0.8 when correlation is high, CD labor supply becomes less predictable and therefore, the model prescribes greater reliance on PD capacity over CD.

[———*Insert Figure 5 Here*———]

3.4.3 Experiment 2: Variability in Undelivered Cost.

In Experiment 2, we include Undelivered Cost, u, as an experimental factor (refer to Figure S4.1 and Table S4.2). We found that all four variables significantly contribute to the variance in Q^*. Demand Uncertainty (CV_D) shows a strong association with Q^* ($F(4) = 2728.9, p < 0.01$), similar to CD Supply Uncertainty (CV_S) ($F(4) = 1674.0, p < 0.01$), CD Supply Correlation (ρ) ($F(2) = 51105.7, p < 0.01$), and Undelivered Cost (u) ($F(2) = 86064.3, p < 0.01$). Additionally, the interactions between two, three, and four variables are also significant, as reported in Table 6.

Notably, the relationship between two-sided uncertainty and Q^* is highly dependent on the Undelivered Cost, u since this factor captures the consequences of insufficient or unreliable delivery capacity. As shown in Figure 6, the direction of the CV_D and Q^* relationship is initially negative but reverses as failed delivery cost increases, across levels of CD Supply Correlation. At lower levels of u, the risk of undelivered costs is low, so the model reduces Q^* to reflect this risk. However, as u increases, the cost of failed deliveries becomes prohibitive, and the model increases Q^* to mitigate this risk. Across levels of CD Supply Correlation, the direction of the CV_S and Q^* relationship is initially positive, but the magnitude of the relationship grows as failed delivery cost increases.

[———*Insert Figure 6 Here*———]

[———*Insert Table 6 Here*———]

4 DISCUSSION AND CONCLUSIONS

Our research reveals how two-sided uncertainty stemming from demand and CD supply uncertainty can affect LMD capacity planning decisions. The research was motivated by *AffordableMeals*, a Brazilian DTC company and their logistics planning challenge that emerged following a strategic shift to lower-priced products, which transformed a manageable DTC activity into a complex LMD capacity planning problem. We have adapted the classic NVP2S framework (Chung and Flynn 2001) into a Monte Carlo simulation optimization model that can systematically address this problem by relaxing two key NVP2S assumptions to better reflect the operational reality of *AffordableMeals*. Applying the model with the company's empirical data, we focused on two important factors (the degree of correlation in CD labor supply across different price tiers and the cost penalties associated with delivery failures) to observe the planning implications of two-sided uncertainty. Our quantitative analyses reveal several theoretical relationships and practical insights about how companies should balance PD and CD capacity under two-sided uncertainty. Our results are empirically supported by *AffordableMeals'* independent evolution toward a similar hybrid delivery fleet strategy.

The model generally prescribes higher PD capacity when demand uncertainty and CD labor supply uncertainty increase. This is unexpected because a company seeking to minimize LMD costs would likely allocate deliveries based on unit delivery costs. In fact, this was *AffordableMeals'* initial decision heuristic, which is depicted in Figure 7. According to their LMD planning manager in our initial interviews, *"... the more [crowdsourcing] deliveries I do, the cheaper, it will cost because [crowdsourcing] is cheaper, it's almost always cheaper than my dedicated vehicles. So, what I'm trying to do with my model is to use as much as [crowdsourcing] as I can."* However, we show that focusing on two-sided uncertainty can reduce total LMD cost rather more than focusing on unit delivery costs alone, a conclusion that *AffordableMeals* reached independently. This comparability of outcome with our results corroborates and validates our research, and is consistent with both newsvendor theory (Chung and Flynn 2001, Ji and Kamrad 2019, Petruzzi et al. 2009) and recent findings on sharing economy platforms (Cachon et al. 2017, Besbes et al. 2021).

[———*Insert Figure 7 Here*———]

We also offer a set of theoretical propositions based on the empirical results. This inductive quantitative research uses the specific case of *AffordableMeals* to propose how two-sided uncertainty, CD labor supply correlation, and failed delivery costs impact LMD capacity planning more generally. The key decision variable of our Monte Carlo simulation optimization model is Q^*, the optimal PD capacity. This variable represents the number of deliveries to allocate to PD capacity that optimizes total system-level costs associated with not just PD and CD across all price tiers, but CD labor availability, reliability, and failed delivery costs as well.

In the *AffordableMeals* context, greater two-sided uncertainty resulted in their need to increase PD capacity across all delivery periods. In fact, six months after the company's strategic shift to offer low-cost products amplified two-sided uncertainty, the leadership team confirmed that they had to increase PD capacity due to CD unreliability despite the generally lower CD delivery costs. This independently-reached outcome aligns with our model's results as shown in the Base Case experiment, suggesting the following theoretical proposition:

Proposition 1 *Higher levels of two-sided uncertainty are associated with higher PD capacity needed to minimize total system LMD costs.*

Second, our experiments uncovered a more nuanced relationship between two-sided uncertainty and optimal PD capacity than previously established (see Figure 4). When CD labor supply uncertainty is high, demand uncertainty exhibits a distinctive U-shaped relationship with PD capacity. Such a nonlinear relationship does not appear at lower levels of CD labor supply uncertainty. This interaction effect between the two sources of uncertainty reveals important dynamics in LMD capacity planning. At low levels of demand uncertainty, the model prescribes higher PD capacity despite predictable demand because high CD labor supply uncertainty creates substantial delivery failure risk. Similarly, at high levels of demand uncertainty, higher PD capacity buffers against both volatile demand and unreliable CD labor supply. However, at moderate levels of demand uncertainty, the model shifts to prescribing greater reliance on CD capacity to take advantage of its lower cost structure, effectively reducing the risk of expensive idle PD capacity. This counterintuitive finding suggests that optimal capacity allocation depends on the complex interplay between both sources of uncertainty rather than either one in isolation. The *AffordableMeals* case supports this finding, as their experience showed a need to carefully balance reliability of PD capacity against the cost advantages of CD when facing varying levels of uncertainty. Therefore, we propose:

Proposition 2 *When CD labor supply uncertainty is high, the relationship between demand uncertainty and* *optimal PD capacity follows a U-shaped curve; when CD labor supply uncertainty is low, the relationship between demand uncertainty and optimal PD capacity is monotonically increasing.*

Third, our experiments revealed how correlation across tiers of CD labor supply affects optimal LMD capacity allocation (see Figure 5). When CD labor supply is highly correlated across all price tiers, such as during weather events affecting all CD drivers similarly or on weekends when social events are taking place, the model prescribes lower PD capacity. This occurs because the predictability of aggregate CD labor supply improves, which reduces the need for PD capacity. However, this predictability diminishes as CD labor supply uncertainty increases. As a result, more PD capacity is required to manage such supply-side volatility. This finding aligns with *AffordableMeals'* observations that certain times (like Saturday evenings) or conditions (rainy weather) affect CD availability across all tiers simultaneously, making CD labor supply more predictable, but less reliable. We thus propose:

Proposition 3 *Increased correlation across CD labor supply tiers reduces the optimal PD capacity needed that minimizes total system LMD costs.*

Fourth, our experiments show how delivery failure costs fundamentally affect capacity decisions under two-sided uncertainty (see Figure 6). When failed delivery costs are low, demand uncertainty exhibits a negative association with PD capacity allocation, as the minimal consequences of a failed delivery reduce the need for additional PD capacity. However, as failed delivery costs rise, the risk of delivery failures becomes more severe, necessitating higher PD capacity to buffer against demand volatility. Increased CD labor supply uncertainty further amplifies the need for PD capacity under high undelivered costs. For example, the model recommends higher PD capacity when failed deliveries could compromise frozen food quality, requiring disposal of returned items and costly redelivery attempts. *AffordableMeals*, which relies on temperature-controlled delivery service, supports this finding in with the observation that failed deliveries often incur product spoilage and disposal costs as well as expensive redelivery costs. Hence, our fourth proposition is:

Proposition 4 *Higher failed delivery costs strengthen the positive relationship between two-sided uncertainty and optimal PD capacity that minimizes total system LMD costs.*

Follow-up meetings with the *AffordableMeals* leadership team six months after their strategic shift in early 2024 provided compelling validation of the model's effectiveness and theoretical propositions. The company had independently evolved toward strategies that aligned with our model's recommendations through trial-and-error experimentation. For instance, after discovering poor CD reliability during night shifts, they doubled their PD capacity during these periods. Our analysis of their data showed this practice aligned with model prescriptions. Given their empirical demand uncertainty ($CV_D = 0.34$) and CD labor supply uncertainty ($CV_S = 0.36$), the model recommends allocating $Q^* =$ 345 deliveries to dedicated capacity – higher than their initial average of 300 deliveries per day during the period covered.

Interestingly, *AffordableMeals* maintained a hybrid delivery model rather than relying exclusively on either PD or CD capacity. Additional Monte Carlo simulation experiments (see Appendix C) comparing total expected costs between PD-only, CD-only, and hybrid fleet confirmed the optimality of this approach. As their last mile logistics manager explained, *"...if I depend only on [crowdsourcing], at a certain point, we won't be able to deliver everything we have to deliver."* The company's real-world experience with capacity planning challenges leads to our fifth and final proposition:

Proposition 5 *Under two-sided uncertainty, a hybrid delivery model combining both PD and CD capacity has lower total expected LMD costs compared to single-source delivery models.*

Collectively, this study advances understanding of LMD capacity planning by theorizing and empirically validating how two-sided uncertainty affects optimal LMD capacity allocation decisions. Our findings extend previous work that considered either demand or CD labor supply uncertainty in isolation (Castillo et al. 2022, Gdowska et al. 2018, Huang and Ardiansyah 2019). Viewing the interaction of demand and CD labor supply uncertainty reveals more complex relationships that must be navigated in last mile capacity planning. *AffordableMeal's* evolution toward strategies aligned with our model's prescriptions provides compelling evidence of its theoretical propositions. The research also provides practical guidance for managers facing similar challenges in LMD capacity planning, particularly those dealing with two-sided uncertainty.

4.1 Theoretical Implications

This research contributes a novel theoretical perspective on LMD to the LSCM literature. While previous research has either assumed demand and CD labor supply uncertainty away, or focused on one or the other in isolation, we examine their interaction and subsequent capacity planning implications. By adapting classic inventory theory (the NVP) to this context, we demonstrate how PD capacity can be viewed as an inventory-like asset with optimal levels that can be optimized to balance delivery cost and service levels.

Adapting Chung and Flynn's (2001) NVP2S requires relaxing two key assumptions that do not affect the reality of last mile capacity planning. First, the assumption that secondary CD capacity is deterministic can be overcome by incorporating stochastic CD labor supply. Second, we accounted for the possibility that secondary CD capacity may be cheaper than PD capacity, contrary to traditional NVP2S assumptions about secondary source costs. These modifications not only extend the classic inventory theory framework to address LMD challenges, but also contribute back to theory by demonstrating how the NVP2S can evolve to remain relevant in modern logistics contexts characterized by two-sided uncertainty.

Our theoretical propositions advance understanding of capacity planning capacity planning under two-sided uncertainty in several ways. First, we establish that the relationship between uncertainty and optimal PD capacity is not uniformly positive, but depends on relative levels of demand and CD labor supply uncertainty (P1, P2). Second, we show how CD labor supply correlation across price tiers affects optimal capacity allocation decisions (P3), introducing a new dimension to capacity planning. Third, we demonstrate how failed delivery costs moderate these relationships (P4), extending previous work on service failure costs to the CD context. Finally, we establish theoretical support for hybrid delivery models under two-sided uncertainty (P5).

4.2 Managerial Implications: The Risk of Minimizing Cost at All Costs

There are several managerial implications for LMD capacity planning stemming from our research. First, and most critically, focusing solely on the lowest unit delivery costs systematically leads to suboptimal decisions. This finding directly contradicts the intuitive managerial heuristic of simply choosing the cheapest delivery option. Table 7 shows expected landed costs of CD across all tiers, calculated as the sum of CD spend divided by CD deliveries. While average expected CD costs appear lower than PD costs, this simple comparison dangerously masks the effect of self-scheduling behavior and two-sided uncertainty. For instance, at high uncertainty levels ($CV_D = 0.8$, $CV_S = 0.8$), our model recommends that *AffordableMeals* allocate more deliveries to the "expensive" PD capacity ($Q^* = 406.4$) despite its higher unit cost (R\$20.5 vs R\$12.6). This counterintuitive result arises because the model minimizes total system costs by buffering against both demand spikes and CD labor unreliability. Complex last mile delivery systems require systematic optimization tools that account for two-sided uncertainty, not simplistic cost heuristics that can mislead.

[———*Insert Table 7 Here*———]

Second, our study demonstrates how historical demand and CD labor supply data can aid managers in deciding the optimal allocation of last mile deliveries to PD capacity. As *AffordableMeal's* experience indicates, relying on basic statistics and managerial judgment can lead to optimal solutions through costly trial-and-error experimentation, but our model provides a systematic approach for optimizing LMD capacity allocation. This is particularly relevant for strategic planning as under- or over-investing in PD capacity will significantly impact a firm's bottom line, as well as its ability to meet service level promises made to its customers.

Third, our findings suggest specific conditions when managers should adjust their PD capacity allocation. When both demand and CD labor supply uncertainty are high, increase PD capacity. During periods of correlated CD labor supply, reduce PD capacity. When failed delivery costs are high, increase PD capacity. At moderate demand uncertainty, reduce PD capacity to reduce idle PD capacity costs.

Finally, *AffordableMeal's* experience provides concrete evidence of these implications in practice. The company adopted a hybrid delivery approach, which emerged from trial-and-error testing and whose outcomes aligned with our model's results. The current practice of adjusting capacity by shift, accounting for predictable patterns in CD reliability, demonstrates how our theoretical insights translate into strategic planning decisions. Companies facing similar challenges can learn from this research how to accommodate two-sided uncertainty in last mile planning and potentially avoid costly experimentation periods.

4.3 Future Research

This study presents several avenues for future research to enhance our understanding of LMD capacity planning under uncertainty. We assumed a neoclassical view of price tiers in the CD labor supply. However, emerging research suggests alternative CD labor supply theories such as income targeting (Camerer et al. 1997, Zha et al. 2018, He et al. 2022, Miao et al. 2023). This theory posits that CD drivers set daily income targets and work only until they meet these targets. Surge pricing might decrease labor supply once drivers reach their income goals. Future research could examine how such behavioral dynamics affect our propositions about last mile capacity planning.

Our model assumes that CD drivers accept tasks based solely on remuneration. However, factors like delivery locations and perceived time-and-effort requirements also influence their decisions (Castillo et al.

2022). In fact, this study focuses on deliveries in a dense urban market; rural delivery contexts could significantly impact how drivers decide to accept task offers. Further, rural contexts have challenging characteristics that impact route efficiency. Future research should incorporate these factors into r_{CD}, the probability of task completion by a CD driver. Investigating how these factors affect r_{CD} and the overall level of two-sided uncertainty could provide deeper insights into delivery capacity planning.

While our model minimizes total expected cost, it does not explicitly address service levels. Future research should explore the practical cost performance[4] of the model and its impact on service quality. When CD drivers reject delivery tasks, creating a backlog can worsen fulfillment times and potentially compromise service levels. Different CD platforms and driver management styles can affect these dynamics. Additionally, demand might be negatively impacted by delivery backlogs, influencing repurchase behavior. Future studies should incorporate service levels into the capacity planning problem to better balance cost and quality.

Our analysis focuses on hybrid delivery options combining PD and CD capacity. However, other sources, such as contracting with third-party logistics (3PL) providers, should be considered since the terms of the contract may differ from that of *AffordableMeals* with its providers. Incorporating dedicated 3PL capacity, depending on the contractual arrangement, could offer additional flexibility[5]. Future research could also explore the implications of varying the number of CD capacity sources and integrating 3PL providers into the model. These future research opportunities can help improve the robustness and applicability of LMD capacity planning models. By considering alternative labor supply theories, additional decision factors, service levels, and broader capacity options, future research can provide more comprehensive and practical solutions for managing delivery operations under uncertainty. Additionally, future deductive research can test the generalizability of our theoretical insights about two-sided uncertainty across different markets, cost structures, and operational contexts to establish broader applicability of the capacity planning framework (McGrath 1981, Stank et al. 2024).

Finally, our model uses historical data to capture operational complexities (distance variability, volume fluctuations, demand-labor supply correlations) in aggregate parameters, which is appropriate for strategic capacity planning decisions on a weekly basis. However, future research could explore whether explicit modeling of these factors provides superior decision support for managers requiring greater granularity. For

[4] See Appendix D for an additional Monte Carlo simulation exploration of total expected cost.

[5] See Appendix E for an example of the model's application when a 3PL is available.

instance, companies operating in markets where service quality directly affects demand patterns or where substantial distance or volume variability exists might benefit from incorporating these factors as additional stochastic parameters. Our Monte Carlo simulation optimization framework is extensible and could accommodate explicit correlation modeling between demand and labor supply uncertainties, distance and volume variability across deliveries, or other operational complexities. Investigating whether such detailed modeling enhances capacity planning insights compared to our historical data approach could help determine optimal levels of model complexity for different managerial contexts and decision requirements.

References

Akeb, H., Moncef, B., and Durand, B. (2018). Building a collaborative solution in dense urban city settings to enhance parcel delivery: An effective crowd model in Paris. *Transportation Research Part E: Logistics and Transportation Review*, 119:223–233.

Allon, G. and Van Mieghem, J. A. (2010). Global Dual Sourcing: Tailored Base-Surge Allocation to Near- and Offshore Production. *Management Science*, 56(1):110–124.

Amaran, S., Sahinidis, N. V., Sharda, B., and Bury, S. J. (2015). Simulation optimization: a review of algorithms and applications. *Annals of Operations Research*, 240(1):351–380.

April, J., Glover, F., Kelly, J. P., and Laguna, M. (2003). Simulation-based optimization: practical introduction to simulation optimization. pages 71–78. Winter Simulation Conference.

Archetti, C., Guerriero, F., and Macrina, G. (2021). The online vehicle routing problem with occasional drivers. *Computers & Operations Research*, 127:105144.

Arrow, K. J., Harris, T., and Marschak, J. (1951). Optimal inventory policy. *Econometrica*, 19(3):250–272.

Arslan, A. M., Agatz, N., Kroon, L., and Zuidwijk, R. (2019). Crowdsourced Delivery—A Dynamic Pickup and Delivery Problem with Ad Hoc Drivers. *Transportation Science*, 53(1):222–235.

Awaysheh, A., Frohlich, M. T., Flynn, B. B., and Flynn, P. J. (2021). To err is human: Exploratory multilevel analysis of supply chain delivery delays. *Journal of Operations Management*, 67(7):882–916.

Baldi, B., Confente, I., Russo, I., and Gaudenzi, B. (2024). Consumer-Centric Supply Chain Management: A Literature Review, Framework, and Research Agenda. *Journal of Business Logistics*, 45(4):e12399.

Behrendt, A., Savelsbergh, M., and Wang, H. (2024). Task assignment, pricing, and capacity planning for a hybrid fleet of centralized and decentralized couriers. *Transportation Research Part C: Emerging Technologies*, 160:104533.

Benjaafar, S. and Hu, M. (2020). Operations Management in the Age of the Sharing Economy: What Is Old and What Is New? *Manufacturing & Service Operations Management*, 22(1):93–101.

Besbes, O., Castro, F., and Lobel, I. (2021). Surge Pricing and Its Spatial Supply Response. *Management Science*, 67(3):1350–1367.

Bisi, A., Dada, M., and Tokdar, S. (2011). A Censored-Data Multiperiod Inventory Problem with Newsvendor Demand Distributions. *Manufacturing & Service Operations Management*, 13(4):525–533.

Buldeo Rai, H., Verlinde, S., Merckx, J., and Macharis, C. (2017). Crowd logistics: an opportunity for more sustainable urban freight transport? *European Transport Research Review*, 9(3):13.

Burke, G. J., Carrillo, J. E., and Vakharia, A. J. (2009). Sourcing Decisions with Stochastic Supplier Reliability and Stochastic Demand. *Production and Operations Management*, 18(4):475–484.

Cachon, G. P., Daniels, K. M., and Lobel, R. (2017). The Role of Surge Pricing on a Service Platform with Self-Scheduling Capacity. *Manufacturing & Service Operations Management*, 19(3):368–384.

Callil, V., Picanço, M., Costanzo, D., Shiraishi, J., Trindade, G., Rumenig, E., and Mata-Machado, D. (2023). Mobilidade urbana e logística de entregas: um panorama sobre o trabalho de motoristas e entregadores com aplicativos. Technical report, CEBRAP, São Paulo, Brazil.

Camerer, C., Babcock, L., Loewenstein, G., and Thaler, R. (1997). Labor Supply of New York City Cabdrivers: One Day at a Time. *The Quarterly Journal of Economics*, 112(2):407–441.

Carbone, V., Rouquet, A., and Roussat, C. (2017). The Rise of Crowd Logistics: A New Way to Co-Create Logistics Value. *Journal of Business Logistics*, 38(4):238–252.

Carson, Y. and Maria, A. (1997). Simulation optimization: methods and applications. pages 118–126. IEEE Computer Society.

Castillo, J. C., Knoepfle, D., and Weyl, E. G. (2024). Matching and Pricing in Ride Hailing: Wild Goose Chases and How to Solve Them. *Management Science*, 0(0).

Castillo, V. E., Bell, J. E., Mollenkopf, D. A., and Stank, T. P. (2021). Hybrid last mile delivery fleets with crowdsourcing: A systems view of managing the cost-service trade-off. *Journal of Business Logistics*, 43(1):36–61.

Castillo, V. E., Bell, J. E., Rose, W. J., and Rodrigues, A. M. (2018). Crowdsourcing Last Mile Delivery: Strategic Implications and Future Research Directions. *Journal of Business Logistics*, 39(1):7–25.

Castillo, V. E., Mollenkopf, D. A., Bell, J. E., and Esper, T. L. (2022). Designing technology for on-demand delivery: The effect of customer tipping on crowdsourced driver behavior and last mile performance. *Journal of Operations Management*, 68(5):424–453.

Cattani, K. D., Dahan, E., and Schmidt, G. M. (2008). Tailored capacity: Speculative and reactive fabrication of fashion goods. *International Journal of Production Economics*, 114(2):416–430.

Chen, C., Pan, S., Wang, Z., and Zhong, R. Y. (2016). Using taxis to collect citywide E-commerce reverse flows: a crowdsourcing solution. *International Journal of Production Research*, 55(7):1833–1844.

Choi, T.-M., editor (2012). *Handbook of Newsvendor Problems: Models, Extensions and Applications*, volume 176 of *International Series in Operations Research & Management Science*. Springer, New York.

Chung, C.-S. and Flynn, J. (2001). A newsboy problem with reactive production. *Computers & Operations Research*, 28(8):751–765.

Cohen, M. C. and Zhang, R. (2022). Competition and coopetition for two-sided platforms. *Production and Operations Management*, 31(5):1997–2014.

Dada, M., Petruzzi, N. C., and Schwarz, L. B. (2007). A Newsvendor's Procurement Problem when Suppliers Are Unreliable. *Manufacturing & Service Operations Management*, 9(1):9–32.

Dantzig, G. B. and Ramser, J. H. (1959). The truck dispatching problem. *Management Science*, 6(1):80–91.

Dayarian, I. and Pazour, J. (2022). Crowdsourced order-fulfillment policies using in-store customers. *Production and Operations Management*, 31(11):4075–4094.

Dayarian, I. and Savelsbergh, M. (2020). Crowdshipping and Same-day Delivery: Employing In-store Customers to Deliver Online Orders. *Production and Operations Management*, 29(9):2153–2174.

Devari, A., Nikolaev, A. G., and He, Q. (2017). Crowdsourcing the last mile delivery of online orders by exploiting the social networks of retail store customers. *Transportation Research Part E: Logistics and Transportation Review*, 105:105–122.

Esper, T. L., Castillo, V. E., Ren, K., Sodero, A., Wan, X., Croxton, K. L., Knemeyer, A. M., DeNunzio, S., Zinn, W., and Goldsby, T. J. (2020). Everything old is new again: The age of consumer-centric supply chain management. *Journal of Business Logistics*, 41(4):286–293.

Fatehi, S. and Wagner, M. R. (2021). Crowdsourcing Last-Mile Deliveries. *Manufacturing & Service Operations Management*, 24(2):791–809.

Garg, N. and Nazerzadeh, H. (2022). Driver Surge Pricing. *Management Science*, 68(5):3219–3235.

Garland, M. (2022). Walmart to purchase 4,500 Canoo electric vehicles to boost last-mile coverage. *Retail Dive*.

Gdowska, K., Viana, A., and Pedroso, J. P. (2018). Stochastic last-mile delivery with crowdshipping. *Transportation Research Procedia*, 30:90–100.

Guda, H. and Subramanian, U. (2019). Your Uber Is Arriving: Managing On-Demand Workers Through Surge Pricing, Forecast Communication, and Worker Incentives. *Management Science*, 65(5):1995–2014.

Guo, X., Lujan Jaramillo, Y. J., Bloemhof-Ruwaard, J., and Claassen, G. (2019). On integrating crowdsourced delivery in last-mile logistics: A simulation study to quantify its feasibility. *Journal of Cleaner Production*, 241:1–13.

He, S., Qiu, L., and Cheng, X. (2022). Surge Pricing and Short-term Wage Elasticity of Labor Supply in Real-Time Ridesharing Markets. *MIS Quarterly*, 46(1):193–228.

Hu, B., Hu, M., and Zhu, H. (2021). Surge Pricing and Two-Sided Temporal Responses in Ride Hailing. *Manufacturing & Service Operations Management*, 24(1):91–109.

Huang, K. and Ardiansyah, M. N. (2019). A decision model for last-mile delivery planning with crowdsourcing integration. *Computers & Industrial Engineering*, 135:898–912.

Janakiraman, G., Seshadri, S., and Sheopuri, A. (2015). Analysis of Tailored Base-Surge Policies in Dual Sourcing Inventory Systems. *Management Science*, 61(7):1547–1561.

Ji, R. and Kamrad, B. (2019). Newsvendor Model as an Exchange Option on Demand and Supply Uncertainty. *Production and Operations Management*, 28(10):2456–2470.

Kafle, N., Zou, B., and Lin, J. (2017). Design and modeling of a crowdsource-enabled system for urban parcel relay and delivery. *Transportation Research Part B: Methodological*, 99:62–82.

Karmarkar, U. S. (1987). The Multilocation Multiperiod Inventory Problem: Bounds and Approximations. *Management Science*, 33(1):86–94.

Khouja, M. (1999). The single-period (news-vendor) problem: literature review and suggestions for future research. *Omega*, 27(5):537–553.

Käki, A., Liesiö, J., Salo, A., and Talluri, S. (2015). Newsvendor decisions under supply uncertainty. *International Journal of Production Research*, 53(5):1544–1560.

Le, T. V., Stathopoulos, A., Van Woensel, T., and Ukkusuri, S. V. (2019). Supply, demand, operations, and management of crowd-shipping services: A review and empirical evidence. *Transportation Research Part C: Emerging Technologies*, 103:83–103.

Lee, K. B., Bellamy, M. A., and Joglekar, N. R. (2022). Distributed service with proximal capacity and pricing on a two-sided sharing economy platform. *Journal of Operations Management*, Early View.

Lin, S., Chen, Y. F., Li, Y., and Shen, Z. M. (2022). Data-Driven Newsvendor Problems Regularized by a Profit Risk Constraint. *Production and Operations Management*, 31(4):1630–1644.

Luy, J., Hiermann, G., and Schiffer, M. (2024). Strategic Workforce Planning in Crowdsourced Delivery With Hybrid Driver Fleets. *Production and Operations Management*, 33(11):2177–2200. Publisher: SAGE Publications.

Ma, L., Xue, W., Zhao, Y., and Zeng, Q. (2016). Loss-averse newsvendor problem with supply risk. *Journal of the Operational Research Society*, 67(2):214–228.

Macrina, G., Di Puglia Pugliese, L., Guerriero, F., and Laporte, G. (2020). Crowd-shipping with time windows and transshipment nodes. *Computers & Operations Research*, 113:1–15.

Masorgo, N., Dobrzykowski, D. D., and Fugate, B. S. (2024). Last-Mile Delivery: A Process View, Framework, and Research Agenda. *Journal of Business Logistics*, 45(4):33.

Masorgo, N., Mir, S., and Hofer, A. R. (2023). You're driving me crazy! How emotions elicited by negative driver behaviors impact customer outcomes in last mile delivery. *Journal of Business Logistics*, 44(4):666–692.

McGrath, J. E. (1981). Dilemmatics: The Study of Research Choices and Dilemmas. *American Behavioral Scientist*, 25(2):179–210. Number: 2 ERIC Number: EJ255600.

Meng, F., Zhou, X., Piao, X., and Chu, D. (2024). Variable neighbourhood search based on Metropolis criterion for crowdsourced delivery scheduling problem in dispatch model. *Computers & Operations Research*, 167:106662.

Merkert, R., Bliemer, M. C. J., and Fayyaz, M. (2022). Consumer preferences for innovative and traditional last-mile parcel delivery. *International Journal of Physical Distribution & Logistics Management*, 52(3):261–284. Publisher: Emerald Publishing Limited.

Miao, W., Deng, Y., Wang, W., Liu, Y., and Tang, C. S. (2023). The effects of surge pricing on driver behavior in the ride-sharing market: Evidence from a quasi-experiment. *Journal of Operations Management*, 69(5):794–822.

Nguyen, D. H., de Leeuw, S., Dullaert, W., and Foubert, B. P. J. (2019). What Is the Right Delivery Option for You? Consumer Preferences for Delivery Attributes in Online Retailing. *Journal of Business Logistics*, 40(4):299–321.

Petruzzi, N. C., Wee, K. E., and Dada, M. (2009). The Newsvendor Model with Consumer Search Costs. *Production and Operations Management*, 18(6):693–704.

Placek, M. (2022). Last mile delivery market size worldwide 2020-2027. *Statista.com*.

Qi, W., Li, L., Liu, S., and Shen, Z.-J. M. (2018). Shared Mobility for Last-Mile Delivery: Design, Operational Prescriptions, and Environmental Impact. *Manufacturing & Service Operations Management*, 20(4):737–751.

Qin, Y., Wang, R., Vakharia, A. J., Chen, Y., and Seref, M. M. (2011). The newsvendor problem: Review and directions for future research. *European Journal of Operational Research*, 213(2):361–374.

Raz, G., Druehl, C. T., and Blass, V. (2013). Design for the Environment: Life-Cycle Approach Using a Newsvendor Model. *Production and Operations Management*, 22(4):940–957.

Reimann, M. (2011). Speculative production and anticipative reservation of reactive capacity by a multi-product newsvendor. *European Journal of Operational Research*, 211(1):35–46.

Repko, M. (2022). Target chases bigger e-commerce profits with new delivery hubs, fleet of drivers. *CNBC*. Accessed 8/2/2022. https://www.cnbc.com/2022/07/25/target-chases-bigger-e-commerce-profits-with-new-delivery-hubs-fleet-of-drivers.html.

Rose, W. J., Bell, J. E., and Griffis, S. E. (2023). Inductive research in last-mile delivery routing: Introducing the

Re-Gifting heuristic. *Journal of Business Logistics*, 44(1):109–140.

Rougès, J.-F. and Montreuil, B. (2014). Crowdsourcing delivery: New interconnected business models to reinvent delivery. In *Towards a smart interconnected era of efficient and sustainable logistics, supply chains, and transportation*.

Saunders, L. W., Castillo, V. E., Rose, W. J., Dohmen, A. E., and Bell, J. E. (2025). Improving Driver Engagement in Delivery and Rideshare Services. *Journal of Business Logistics*, 46(2):27.

Simoni, M. D., Marcucci, E., Gatta, V., and Claudel, C. G. (2020). Potential last-mile impacts of crowdshipping services: a simulation-based evaluation. *Transportation*, 47(4):1933–1954.

Snoeck, A., Merchán, D., and Winkenbach, M. (2020). Revenue management in last-mile delivery: state-of-the-art and future research directions. *Transportation Research Procedia*, 46:109–116.

Stank, T., Saunders, L. W., Scott, A., Autry, C. W., and Esper, T. L. (2024). "Theory will take you only so far" (Nolan, 2023): In search of greater insight through quantitative, observation-based research. *Journal of Business Logistics*, 45(3):e12383.

Straight, B. (2022). Retail leaders: Same-day delivery is no longer optional. *FreightWaves*. Access date: 8/2/2022. https://www.freightwaves.com/news/retail-leaders-same-day-delivery-is-no-longer-an-option.

Suzuki, Y., Chen, H., and Faires, C. (2025). Response Strategies for Demand Surges in Last-Mile Logistics: Managing Delivery Efficiency in Volatile Environments. *Journal of Business Logistics*, 46(1):e70004. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/jbl.70004.

Ta, H., Esper, T. L., and Hofer, A. R. (2018). Designing crowdsourced delivery systems: The effect of driver disclosure and ethnic similarity. *Journal of Operations Management*, 60(1):19–33.

Ta, H., Esper, T. L., Hofer, A. R., and Sodero, A. C. (2025). Reconceptualizing E-Logistics Service Quality (E-LSQ) in Emerging Contexts: The Case of Crowdsourced Delivery. *Journal of Business Logistics*, 46(1).

Ta, H., Esper, T. L., Rossiter Hofer, A., and Sodero, A. (2023). Crowdsourced delivery and customer assessments of e-Logistics Service Quality: An appraisal theory perspective. *Journal of Business Logistics*, 44(3):345–368.

Tokar, T., Williams, B. D., and Fugate, B. S. (2020). I Heart Logistics—Just Don't Ask Me to Pay For It: Online Shopper Behavior in Response to a Delivery Carrier Upgrade and Subsequent Shipping Charge Increase. *Journal of Business Logistics*, 41(3):182–205.

Wang, L., Rabinovich, E., and Guda, H. (2022). An analysis of operating efficiency and policy implications in last-mile transportation following Amazon's integration. *Journal of Operations Management*, Early View. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/joom.1172.

Wang, Y.-J., Wang, Y., Huang, G. Q., and Lin, C. (2024). Public acceptance of crowdsourced delivery from a customer perspective. *European Journal of Operational Research*, 317(3):793–805.

Winkenbach, M. and Janjevic, M. (2018). Classification of Last-Mile Delivery Models for e-Commerce Distribution: A Global Perspective. In *City Logistics 1: New Opportunities and Challenges*, pages 209–229. Wiley, London.

Xu, Y., Lu, B., Ghose, A., Dai, H., and Zhou, W. (2023). The Interplay of Earnings, Ratings, and Penalties on Sharing Platforms: An Empirical Investigation. *Management Science*, 69(10):6128–6146.

Yang, D., Hyland, M. F., and Jayakrishnan, R. (2024). Tackling the crowdsourced shared-trip delivery problem at scale with a novel decomposition heuristic. *Transportation Research Part E: Logistics and Transportation Review*, 188:31.

Zha, L., Yin, Y., and Du, Y. (2018). Surge pricing and labor supply in the ride-sourcing market. *Transportation Research Part B: Methodological*, 117:708–722.

Zhang, B. and Du, S. (2010). Multi-product newsboy problem with limited capacity and outsourcing. *European Journal of Operational Research*, 202(1):107–113.

Zhang, G. (2010). The multi-product newsboy problem with supplier quantity discounts and a budget constraint. *European Journal of Operational Research*, 206(2):350–360.

Zhang, M., Xia, Y., Li, S., Wu, W., and Wang, S. (2019). Crowd Logistics Platform's Informative Support to Logistics Performance: Scale Development and Empirical Examination. *Sustainability*, 11(2):451.

Zhang, Y. and Siemsen, E. (2019). A Meta-Analysis of Newsvendor Experiments: Revisiting the Pull-to-Center Asymmetry. *Production and Operations Management*, 28(1):140–156.

Tables

Table 1: Notations Used in Problem Description

Let...	D represent delivery demand
	a represent the PD capacity unit delivery cost
	l represent the labor cost component of a
	v represent the operating cost component of a
	n represent the number of CD labor supply tiers available
	B_i represent the CD labor supply (measured as the number of deliveries) available at tier $i \in N = \{1, 2, \ldots, n\}$
	ρ represent CD labor supply correlation between tiers $i \in N = \{1, 2, \ldots, n\}$
	b_i represent the cost of CD at price tier $i \in N = \{1, 2, \ldots, n\}$
	r_{CD} represent the reliability of CD drivers in completing a delivery
	r_{priv} represent the reliability of PD drivers in completing a delivery
	u represent the failed delivery cost
	Q^* represent the cost-minimizing number of deliveries to allocate to PD capacity

Table 2: Average Per Unit Delivery Costs for Different CD Tiers

Low Cost Motorcycle	Car	Van	Premium Rate Motorcycle	Super Premium Motorcycle
R$8.8	R$10.8	R$17.3	R$22.0	R$28.7

Note: The CD platforms advertise a base fare determined by distance and piece rates for additional stops (e.g., Low Cost Motorcycle delivery is R$6 per km for 0-1 km, R$1.15 for >1 km and R$1.10 per additional stop). The figures in this table reflect the average landed costs for each tier of CD capacity based on typical delivery distances in the urban market studied.

Table 3: Modeling Process

for Number of iterations
  Create *Demand Uncertainty* based on $D \sim Norm(\mu, \sigma)$
  Create *CD Labor Supply Uncertainty* based on b_i, ρ, and $B \sim Beta(\alpha, \beta)$
  Calculate Q^*:
    Calculate $H(Q)$ based on D, B, a, v, u, b_i, r
    Using binary search, find the smallest Q that satisfies $H(Q) \geq \frac{u-a}{u-v}$
  Calculate Total Cost, $G(Q, D, B)$
  Record iteration result
end for

Table 4: Experimental Scenarios

Scenario	Experimental Variables	Values
Base Case	Demand Uncertainty	CV_D = {0.0, 0.2, 0.4, 0.6, 0.8}
	CD Supply Uncertainty	CV_S = {0.0, 0.2, 0.4, 0.6, 0.8}
Experiment 1 - CD Labor	Demand Uncertainty	CV_D = {0.0, 0.2, 0.4, 0.6, 0.8}
Supply Correlation of B_i	CD Supply Uncertainty	CV_S = {0.0, 0.2, 0.4, 0.6, 0.8}
	CD Supply Correlation	ρ = {0.0, 0.5, 1.0}
Experiment 2 -	Demand Uncertainty	CV_D = {0.0, 0.2, 0.4, 0.6, 0.8}
Changes in	CD Supply Uncertainty	CV_S = {0.0, 0.2, 0.4, 0.6, 0.8}
Undelivered Cost u	CD Supply Correlation	ρ = {0.0, 0.5, 1.0}
	Undelivered Cost	$u = \{\text{R\$28.7}, \text{R\$35.1}, \text{R\$41.6}\}$

Table 5: Empirical Data from *AffordableMeals*

Notation	Description	Value (Unit of Measure)
D	Delivery demand mean	759.9 (deliveries)
a	PD unit delivery cost	20.3 (R$)
l	Labor cost component of a	2.8 (R$)
v	Operating cost component of a	17.6 (R$)
n	Number of CD labor supply tiers available	5 (tiers)
B_i	CD labor supply at tier $i \in N = \{1, 2, \ldots, n\}$	{16.5, 389.8, 9.0, 67.5, 4.2} (deliveries)
ρ	CD labor supply correlation between tiers $i \in N = \{1, 2, \ldots, n\}$	{0.0, 0.5, 1.0} (unitless)
b_i	Cost of CD for tier $i \in N = \{1, 2, \ldots, n\}$	{8.8, 10.8, 17.3, 22.0, 28.7} (R$)
r_{CD}	Reliability of CD drivers in completing a delivery	98.1 (%)
r_{priv}	Reliability of PD drivers in completing a delivery	98.6 (%)
u	Failed Delivery cost	35.1 (R$)

Table 6: Full Factorial ANOVA for Experiment 2

Source of Variation	DF	F-statistic	Pr(>F)
CV_D	4	2056.5	< 0.01
CV_S	4	5540.4	< 0.01
ρ	2	92062.5	< 0.01
u	3	4752.7	< 0.01
$CV_D \times CV_S$	16	9.8	< 0.01
$CV_D \times \rho$	8	184.1	< 0.01
$CV_S \times \rho$	8	1594.0	< 0.01
$CV_D \times u$	12	747.8	< 0.01
$CV_S \times u$	12	3.0	< 0.01
$\rho \times u$	6	673.8	< 0.01
$CV_D \times CV_S \times \rho$	32	16.2	< 0.01
$CV_D \times CV_S \times u$	48	2.1	< 0.01
$CV_D \times \rho \times u$	24	84.9	< 0.01
$CV_S \times \rho \times u$	24	17.3	< 0.01
$CV_D \times CV_S \times \rho \times u$	96	3.1	< 0.01

Table 7: Selected Unit Delivery Costs and Optimal PD Capacity in Base Case

CV_D	CV_S	Expected PD Unit Cost	Avg. Expected CD Unit Cost	Avg. Q^*
0.2	0.2	R$20.5	R$11.3	345.0
0.2	0.8	R$20.5	R$12.9	371.7
0.8	0.2	R$20.5	R$11.2	416.9
0.8	0.8	R$20.5	R$12.6	406.4

Figures

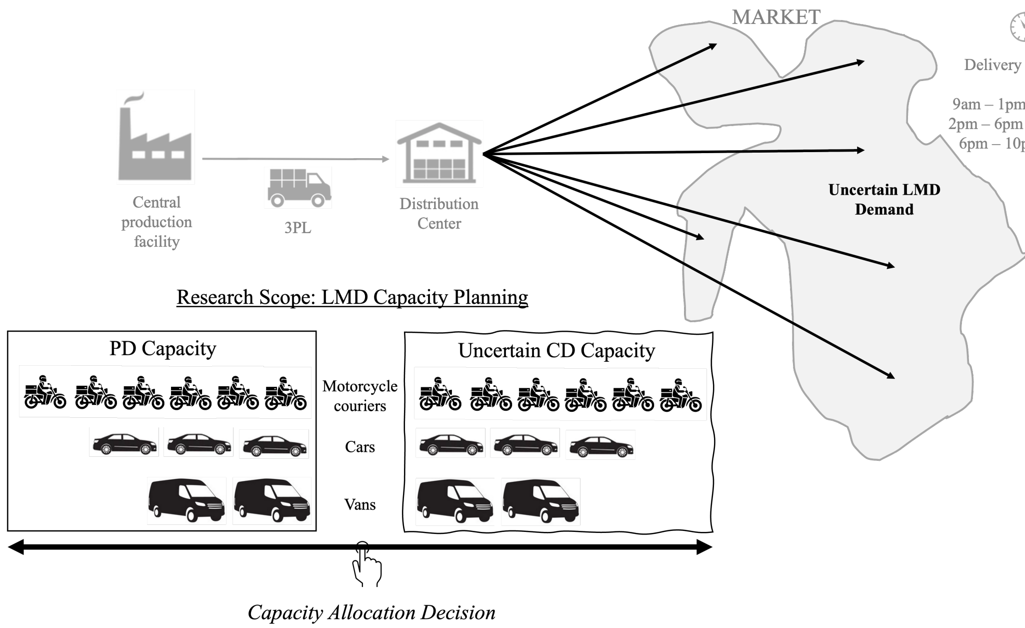

Figure 1: LMD capacity planning schematic based on *AffordableMeals'* operations, which handle an average of 760 daily deliveries across three time windows with both PD and CD. The slider represents the key decision of balancing PD and CD capacity to minimize total expected costs.

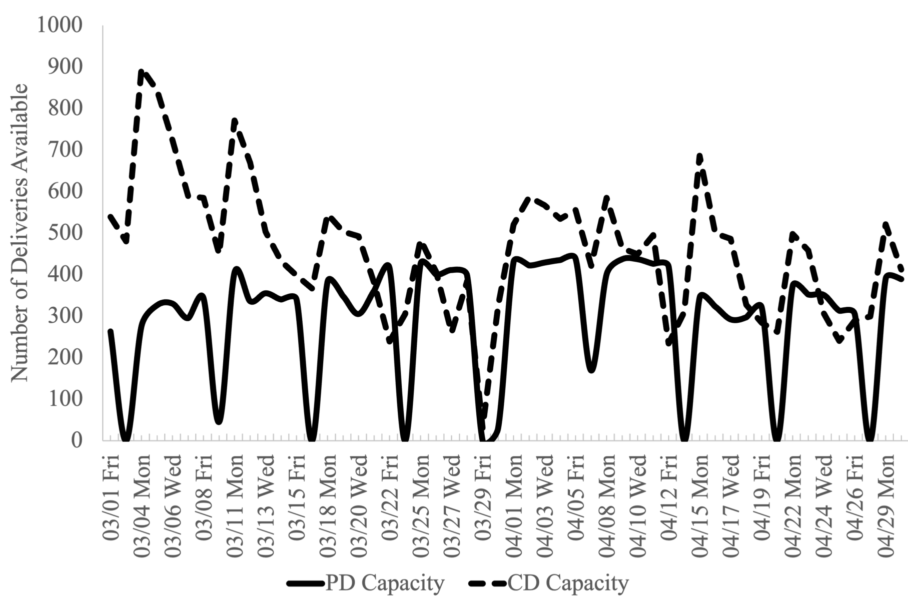

Figure 2: CD capacity shows higher volatility than PD in *AffordableMeals'* LMD operations post-strategic shift.

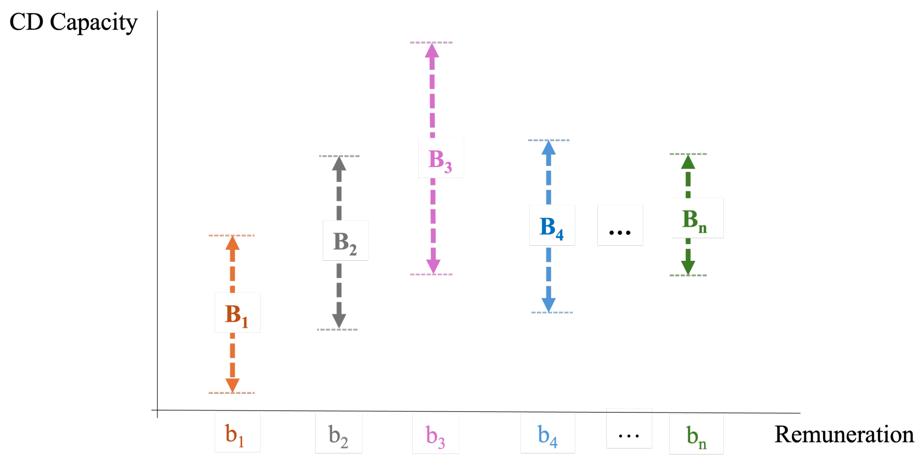

Figure 3: Stochastic CD Labor Supply Model with Price-Dependent Capacity. ***Note: at each price tier b_i, a capacity of B_i is available. The dotted vertical arrows indicates the uncertainty in capacity at each price tier. When CD labor supply capacity at b_i is exhausted, the company must increase to tier b_{i+1}.***

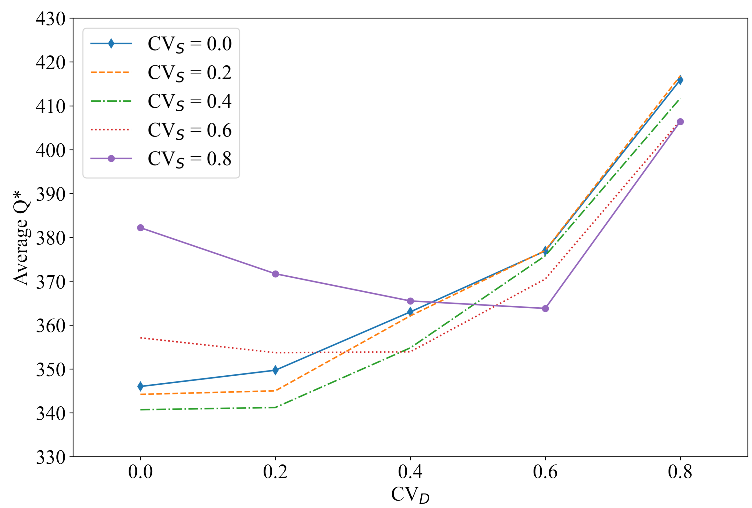

Figure 4: Base Case Scenario Results (5x5 Factorial)

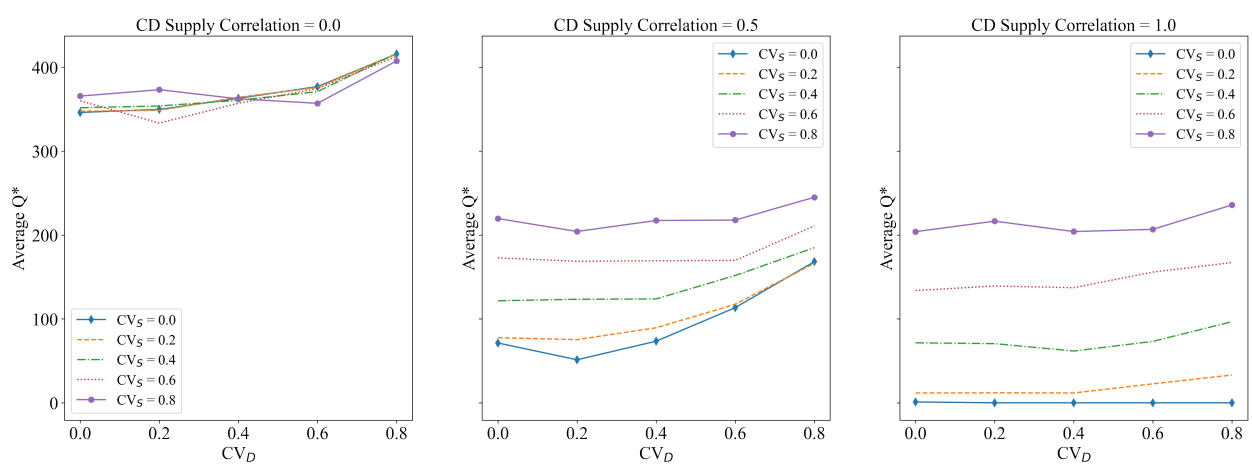

Figure 5: Experiment 1 Results (5x5x3 Factorial)

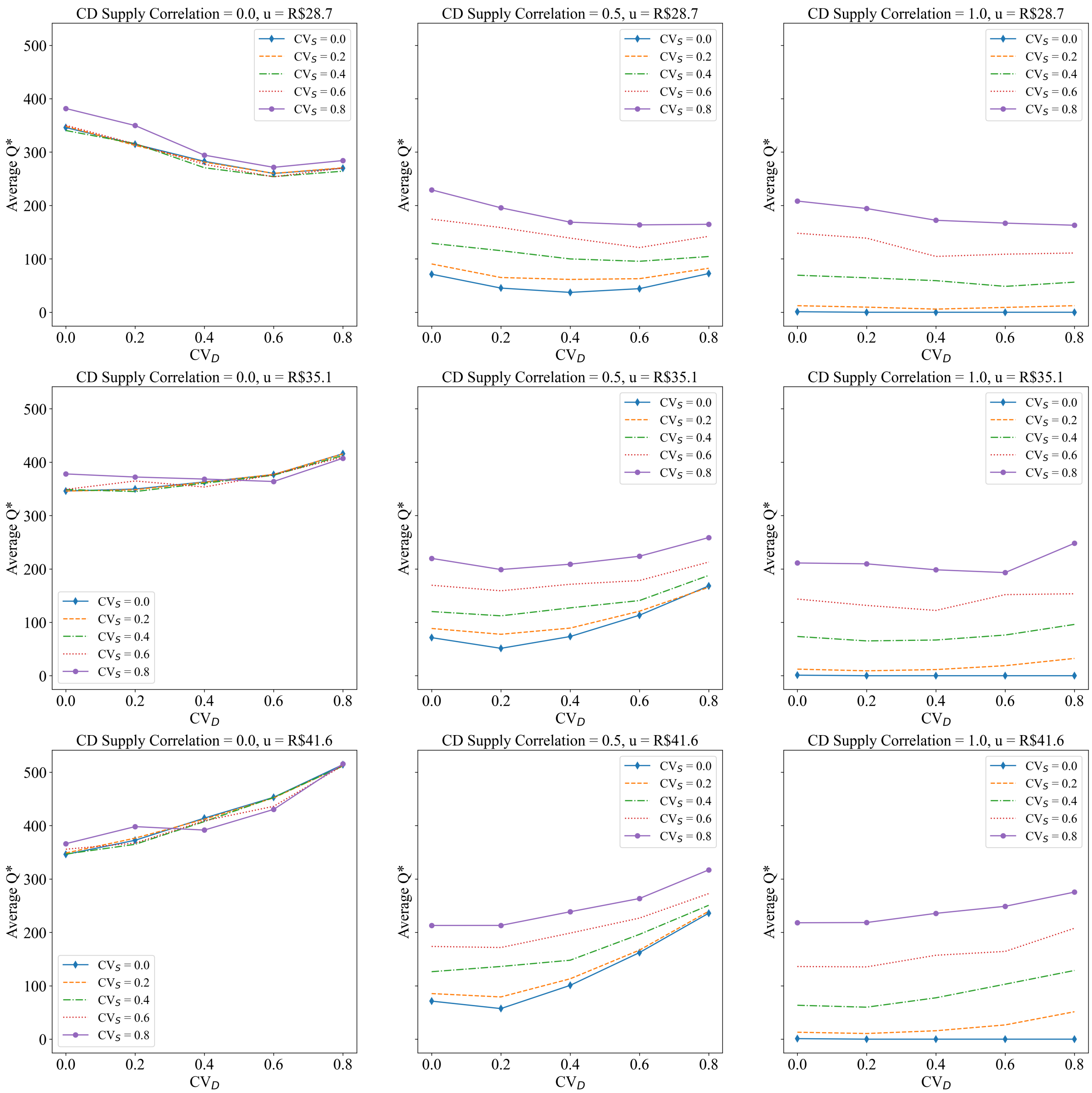

Figure 6: Experiment 2 Results (5x5x3x3 Factorial)

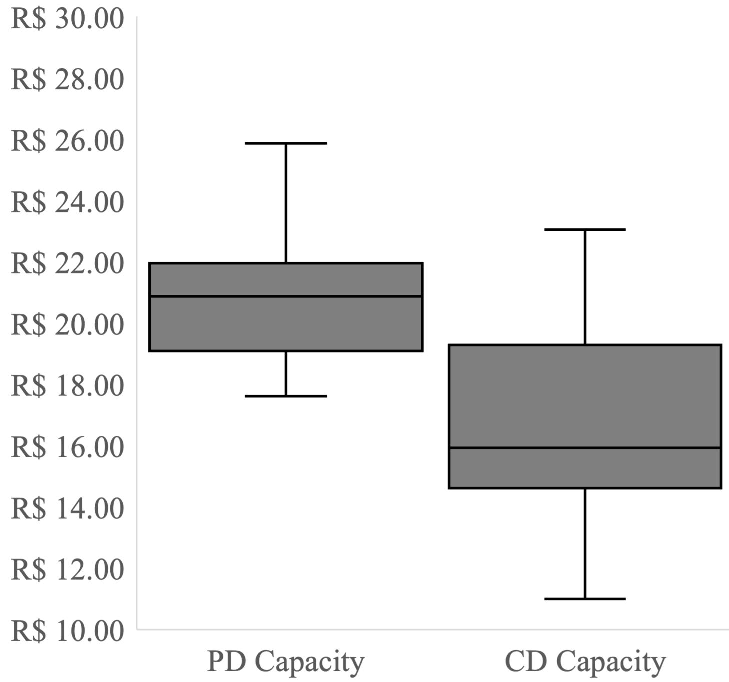

Figure 7: ***AffordableMeals'*** **Average Delivery Costs between PD and CD. Focusing solely on lower delivery cost without accounting for two-sided uncertainty leads to suboptimal decisions.**

Appendix S1 - Mathematical Formulation of the Adapted NVP2S

LMD capacity planning considers both PD and CD sources of capacity, as illustrated by *AffordableMeals'* use of fixed PD capacity per period, supplemented by CD capacity. The incorporation of CD into LMD requires a model that accounts for dual sourcing. As such, we adapt Chung and Flynn's (2001) NVP2S. In the NVP2S model, demand for a single product is satisfied by inexpensive anticipatory production in the first stage. Once actual demand is known in the second stage, a set of more expensive production methods is used to fulfill excess demand. Our LMD capacity planning problem is similar to the NVP2S, in that PD capacity corresponds to anticipatory capacity, while CD capacity corresponds to reactive capacity.

Two assumptions in the original NVP2S would be violated when using it to plan LMD capacity when CD is available. First, the NVP2S assumes deterministic secondary production. As a result, stochastic CD labor supply cannot be incorporated into the NVP2S, making it less than realistic in our research setting. Notably, relaxing this assumption results in a model that is not easily transformable or generalizable due to introducing a nested convolution of random distributions in B_i along with the random distribution D. Further, correlations of the distributions in B_i may amplify the mathematical complexity, raising the question of whether a closed-form solution to this problem exists. We approach the problem by converting the NVP2S to a Monte Carlo simulation and optimization model to accommodate CD labor supply uncertainty.

Second, the original NVP2S model assumes that all first-stage capacity is consumed before any secondary capacity is used since the former is cheaper than the latter. However, when incorporating CD, there may be situations where the cost of CD delivery, b_i, may be less than the operating cost, v, that is, $b_i < v$. This matches the *AffordableMeals'* experience where low-cost CD motorcycles are preferred for smaller order volumes. This implies a possibility where CD labor supply at i is less expensive than a PD driver. Therefore, the original NVP2S model has to be adapted to account for this situation, which we accomplish through a data transformation. Let $w = \text{argmax}\{i \in N \mid b_i < v\}$, and let $D = \max\{\tilde{D} - \sum_{i=1}^{w} B_i,\ 0\}$, where $\tilde{D}$ is the original random variable of demand. The new random variable D, which results from transformation of the $\tilde{D}$ and B_i random variables, is the remaining demand after using the inexpensive CD labor supply $1,2,\ldots,w$. Furthermore, the tiers of CD labor supply $1,2,\ldots,w$ are removed from the problem, that is $N := N \setminus \{1,2,\ldots,w\}$. The result is a new problem that still satisfies the original NVP2S requirement that all original capacity is used before any secondary capacity. We have thus adapted the original NVP2S model to partition demand and CD labor supply when $b_i < v$, such that the portion of demand serviced by the inexpensive CD drivers (that is, $\tilde{D} - D$) is accounted for. Then, the remaining demand D and CD labor supply tiers are treated as a new instance of the NVP2S.

We replace the production cost in the NVP2S model with the cost of a single delivery. Let a denote the first-stage unit cost in the NVP2S model. In our delivery capacity planning model, each delivery incurs a labor cost l, and a operating cost, v. To model the delivery cost using PD capacity, we set $a = l + v$. An idle driver in the PD capacity will still incur the labor cost l but not the operating cost v. Thus, the model accounts for the operating cost savings v, associated with an idle PD driver as a salvage value. To ensure that the problem has a non-trivial solution, it is assumed that $a < u$. Because a PD driver does not complete a delivery in a $1 - r_{priv}$ fraction of the time, the expected cost of PD capacity is

$$\bar{a} = r_{priv}a + (1 - r_{priv})u.$$

In the NVP2S, there are $n = |N|$ secondary production methods, and each has a unit production cost of b_i for $i \in N$. There are no production failures from any of the secondary production methods in the NVP2S, thus the total expected cost in the NVP2S is a function of only unit production costs and total production volume. In our setting, there is a possibility of a failed delivery from the CD tiers. Thus, the expected cost of a capacity level $i \in N$ must account for undelivered cost u. The optimal decision for the company is to not make the delivery i if $b_i > u$ for some $i \in N$. As a result, we assume that $b_i \leq u$ for all $i \in N$ to prevent such trivial outcomes. Because a CD driver at level i does not make a delivery in a $1 - r_{CD}$ fraction of the time, the expected delivery cost of using a driver at CD capacity level $i \in N$ is

$$\bar{b}_i = r_{CD}b_i + (1 - r_{CD})u.$$

The expected costs for each CD labor supply tier are ordered such that $\bar{b}_1 < \bar{b}_2 < \cdots < \bar{b}_n$. If two CD capacity tiers have the same unit delivery cost, that is, if $b_i = b_{i+1}$, they are combined into one larger group.

As soon as Q, the number of LMDs to make with PD capacity, is determined and values of D and $B = (B_1, B_2, \ldots, B_n)$ are observed, the optimal use of CD is determined by crowdsourcing $\min\{\max\{D - Q,\ 0\},\ \sum_{j=1}^{n} B_j\}$ deliveries. Because $\bar{b}_1 < \bar{b}_2 < \cdots < \bar{b}_n$, the retailer does not acquire deliveries at CD labor supply tier i until all of the deliveries at $i - 1$ are exhausted, for $i = 2, 3, \ldots, n$.

Let $\bar{b}_0 = v$ and $\bar{b}_{n+1} = u$. Then, the total cost associated with a decision to allocate Q deliveries to PD capacity is

$$G(Q,D,B) = \bar{a}Q + (D-Q)v + \sum_{i=0}^{n} (\bar{b}_{i+1} - \bar{b}_i) \max\{D - Q - \sum_{j=1}^{i} B_j, 0\}, \quad Q \in I\!R_+.$$

Observe that $G(Q,D,B)$ is convex and piecewise linear, with breakpoints at $B_1, B_2, \ldots, B_n$. Since the goal is to minimize the expected total delivery cost, the objective function of the LMD capacity planning model is

$$\min E_{D,B}[G(Q,D,B)] = \bar{a}Q + E_{D,B}\left[(D-Q)v + \sum_{i=0}^{n} (\bar{b}_{i+1} - \bar{b}_i) \max\{D - Q - \sum_{j=1}^{i} B_j, 0\}\right]. \tag{1}$$

Let $F(D)$ be the cumulative distribution function for the random variable D. For a specified B, let

$$H(Q) = \sum_{i=0}^{n} \frac{(\bar{b}_{i+1} - \bar{b}_i) F(Q + \sum_{j=1}^{i} B_j)}{u - v}.$$

When the expectation of (1) is only over D and not B, Chung and Flynn (2001) show that the optimal number of deliveries to allocate to the PD capacity is Q^*, where

$$Q^* = \min\{Q \in I\!R_+ \mid H(Q) \geq \frac{u-a}{u-v}\}.$$

To incorporate the uncertainty in B and find the expected value of Eq. (1) over both D and B, we implement this optimization model in a Monte Carlo simulation framework that allows us to explore the impact of various distributions of B with varying CD labor supply uncertainties alongside varying demand uncertainties on Q^*.

Appendix S2 - CD Labor Supply Distribution Parameters

To model CD Supply Correlation and the fact that CD labor supply is bounded, B_i is generated from two Beta distributions, which are bounded probability distributions on the interval $[0,1]$. Equation 2 is used to generate B_i.

$$B_i = (1-\rho) \cdot \mathrm{Beta}(\alpha_i, \beta_i) + \rho \cdot \mathrm{Beta}(\alpha_{\text{common}}, \beta_{\text{common}}) \tag{2}$$

The first term is a unique Beta distribution, which generates an independent component of B_i for each of the n capacity tiers. The second term of B_i is generated from a "Common CD Supply Distribution", a Beta distribution common to all n CD labor supply tiers. The parameters α and β for both Beta distributions are listed in Table S2.1.

The portion of the first and second Beta distributions used to determine B_i is regulated by ρ. We set $\rho = \{0.0, 0.5, 1.0\}$, where a $\rho = 0.0$ means that there is no correlation across n and each amount of B_i CD labor supply is unique. Further, $\rho = 1.0$ means there is the same capacity available for each CD labor supply tier.

The α and β shape parameters presented in Table S2.1 were calculated to provide the desired mean and standard deviations needed for the specific coefficient of variation called for in the experimental design. The mean and standard deviation of a Beta distribution are given by:

$$\mu = \frac{\alpha}{\alpha+\beta} \qquad \sigma = \sqrt{\frac{\alpha \cdot \beta}{(\alpha+\beta)^2 \cdot (\alpha+\beta+1)}}$$

Rearranging the mean and standard deviation formulas to solve for α and β:

$$\alpha = \left(\frac{1-\mu}{\sigma^2} - \frac{1}{\mu}\right) \cdot \mu^2 \qquad \beta = \alpha \cdot \left(\frac{1}{\mu} - 1\right)$$

These formulas are used to determine the shape parameters of the Beta distributions that govern CD Labor Supply. We use the empirical capacity means for each tier of CD capacity (divided by 1000 since Beta distributions are bounded on the interval $[0,1]$) and calculate the unique α and β shape parameters needed for each B_i calculation. The resulting value is then multiplied by 1000 to achieve a capacity value commensurate with the empirical data provided by *AffordableMeals*. This process was verified using traces and logging. The mean of the Common CD Supply Distribution uses a weighted average of the empirical capacity of B_i.

Appendix S3 - Monte Carlo Experiments Comparing Delivery Alternatives

This research explores LMD capacity planning when CD is available. Adapting a classic inventory planning model (Chung and Flynn's (2001) two-stage newsvendor problem), our goal is to identify an optimal PD capacity that minimizes total delivery cost. This hybrid approach contrasts with two alternative single-source

Table S2.1: CD Labor Supply Uncertainty Distribution Parameters

	B_1	B_2	B_3	B_4	B_5	B_{common}
$\mu \div 1000$	0.0165	0.3898	0.0090	0.0675	0.0420	0.3221
CVS	$\sigma_1 \div 1000$	$\sigma_2 \div 1000$	$\sigma_3 \div 1000$	$\sigma_4 \div 1000$	$\sigma_5 \div 1000$	$\sigma_{\text{common}} \div 1000$
0.0	0.00	0.00	0.00	0.00	0.00	0.00
0.2	0.00	0.08	0.00	0.01	0.01	0.06
0.4	0.01	0.16	0.00	0.03	0.02	0.13
0.6	0.01	0.23	0.01	0.04	0.03	0.19
0.8	0.01	0.31	0.01	0.05	0.03	0.26
CVS	α_1	α_2	α_3	α_4	α_5	α_{common}
0.0	983499.98	610229.61	990999.99	932519.93	957999.96	677902.76
0.2	24.57	14.87	24.77	23.25	23.91	16.63
0.4	6.13	3.42	6.18	5.76	5.95	3.91
0.6	2.72	1.31	2.74	2.52	2.62	1.56
0.8	1.52	0.56	1.54	1.39	1.45	0.74
CVS	β_1	β_2	β_3	β_4	β_5	β_{common}
0.0	58622559.62	955385.01	109120110.12	12886684.76	21851522.85	1426751.86
0.2	1464.58	23.27	2727.01	321.23	545.33	34.99
0.4	365.41	5.36	681.01	79.61	135.61	8.24
0.6	161.86	2.04	302.12	34.86	59.74	3.29
0.8	90.61	0.88	169.51	19.20	33.19	1.55

delivery options: fully PD capacity and fully CD capacity. To compare these three alternatives, we construct a traditional Monte Carlo experiment, simulating the performance of each option under *AffordableMeals'* empirical setting. We propose that lowest total expected LMD cost can be obtained through a hybrid option rather than single-source options since two-sided uncertainty in demand, CV_D, and CD labor supply, CV_S introduce real-world characteristics that complicated LMD capacity planning.

The experimental approach in this section has two phases. First, the NVP2S developed in the main study is run using a set of empirical parameters. The result is a Q^* that prescribes the number of deliveries to allocate to PD capacity. This is the same process as in the main analysis. Second, in the Monte Carlo experiment phase, we simulate LMD demand and estimate total expected costs using the hybrid option based on the Q^* obtained in the first phase.

In the latter phase, we simulate one day of delivery demand based on the empirical data from *AffordableMeals* ($\mu = 759.9, \sigma = 254.8$), which results in $CV_D = 0.34$. Focusing on the empirical demand (as opposed to including CV_D as an experimental factor) allows for understanding what *AffordableMeals* could experience when implementing this model. The experimental design is, therefore, a 5x3 factorial with five levels of CD Labor Supply Uncertainty and three levels of CD Labor Supply Correlation. The parameters employed are adopted from the main analysis as shown in Table S3.1. Each cell of the experimental design is replicated 1000 times and we calculate the corresponding total LMD cost. We then compare the total costs of the hybrid option to those of the PD-only and CD-only alternatives.

Table S3.1: Monte Carlo Experiment Parameters

Parameter	Value
CD Labor Supply Uncertainty	$CV_S = \{0.0, 0.2, 0.4, 0.6, 0.8\}$
CD Labor Supply Correlation	ρ = {0.0, 0.5, 1.0}
PD Unit Delivery Cost	$a =$ R\$20.3
CD Unit Delivery Cost	$b_i = \{$R\$8.8, R\$10.8, R\$17.3, R\$22.0R\$28.7$\}$
CD Labor Supply Distributions at each tier	Depends on CV_S; see Table S2.1
Failed delivery Cost	$u =$ R\$35.1

The results of the Monte Carlo experiment are visualized in Figure S3.1. The most prominent finding is that the hybrid option results in lower total LMD cost than the PD-only option. In fact, PD-only is the most expensive of the three options. This is because the PD capacity is more expensive ($a =$ R\$20.3) than the majority of the CD capacity available. For $i = 1, 2, 3$, the costs of b_i are R\$8.8, R\$10.8, and R\$17.3 respectively, where $b_i < a$ for each i. Additionally, the sum of b_1, b_2, and b_3 is $\bar{B}_i = 415.3$ deliveries, so approximately half (54.6%) of the mean demand can be served with CD at a lower cost than PD.

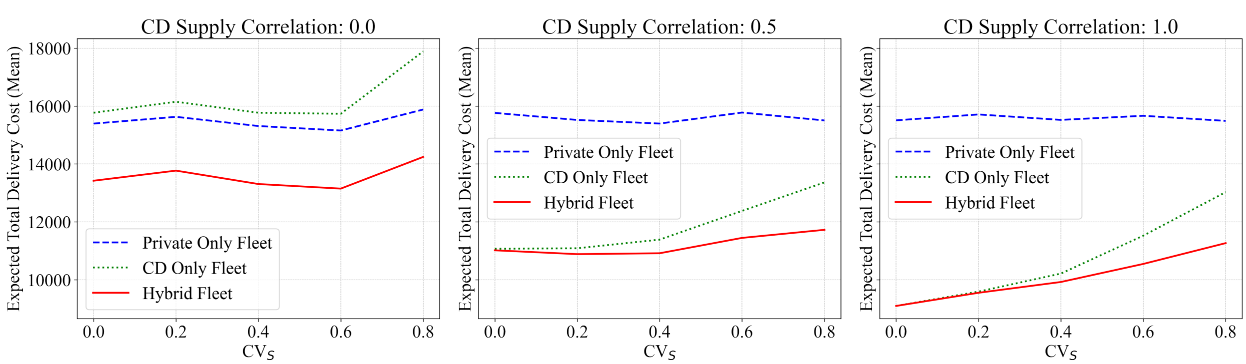

Figure S3.1: Monte Carlo Experiment Results

Appendix S4 - Additional Simulation Analysis

This appendix provides an additional simulation analysis to further explore the impact of two-sided uncertainty on optimal private capacity under an additional contextual variable: private unit delivery cost variability. Table S4.1 summarizes the factorial design of this experiment.

Table S4.1: Additional Experimental Scenario

Scenario	Experimental Variables	Values
Experiment 3 -	Demand Uncertainty	CV_D = {0.0, 0.2, 0.4, 0.6, 0.8}
Changes in PD	CD Supply Uncertainty	CV_S = {0.0, 0.2, 0.4, 0.6, 0.8}
Unit Delivery Cost a	CD Supply Correlation	ρ = {0.0, 0.5, 1.0}
	PD Unit Delivery Cost	$a = \{\text{R\$6.6}, \text{R\$20.3}, \text{R\$34.0}\}$

D.1 Experiment 3: PD Unit Delivery Cost Variability.

Experiment 3 adds *PD Unit Delivery Cost*, $a = \{\text{R\$6.6}, \text{R\$20.3}, \text{R\$34.0}\}$, as an additional experimental factor. This enables us to consider how variability in PD cost relative to CD cost impacts Q^*. In the Base Case, and Experiments 1 and 2, $a =$ R\$20.3 per delivery, which is calculated from the empirical data.

Figure S4.1 depicts the results of Experiment 3, in which PD Unit Delivery Cost, a, is added as an experimental factor. As shown in Table S4.2, the direct associations with Q^* of all four experimental variables are significant. PD Unit Delivery Cost, a, explains the most amount of variance in Q^*, which conveys the importance of monitoring a relative to CD costs. When $a < b_1$, the model prescribes large PD capacity to take advantage of low PD unit delivery costs. Conversely, when $a > b_4$, no PD capacity is prescribed because it is less expensive to use CD drivers to satisfy demand.

The two-way interactions are significant as well, with the $CV_D \times a$ interaction explaining the most amount variance in Q^*. The interaction shows that as demand uncertainty increases (i.e., it becomes more volatile), the optimal PD capacity Q^* is mostly impacted by a. The next interaction that significantly explains a large amount of variance in Q^* is $\rho \times a$, which shows a similar relationship: as correlation of CD Supply Uncertainty increases, the optimal PD capacity Q^* depends upon PD unit delivery cost. The three-way interactions, along with the four-way interaction, are also significant. These results reinforce the stabilizing effect of CD Supply Correlation and subsequent need for less PD capacity. Conditions that increase the correlation across CD labor supply tiers along with changes in a induce a change in the optimal

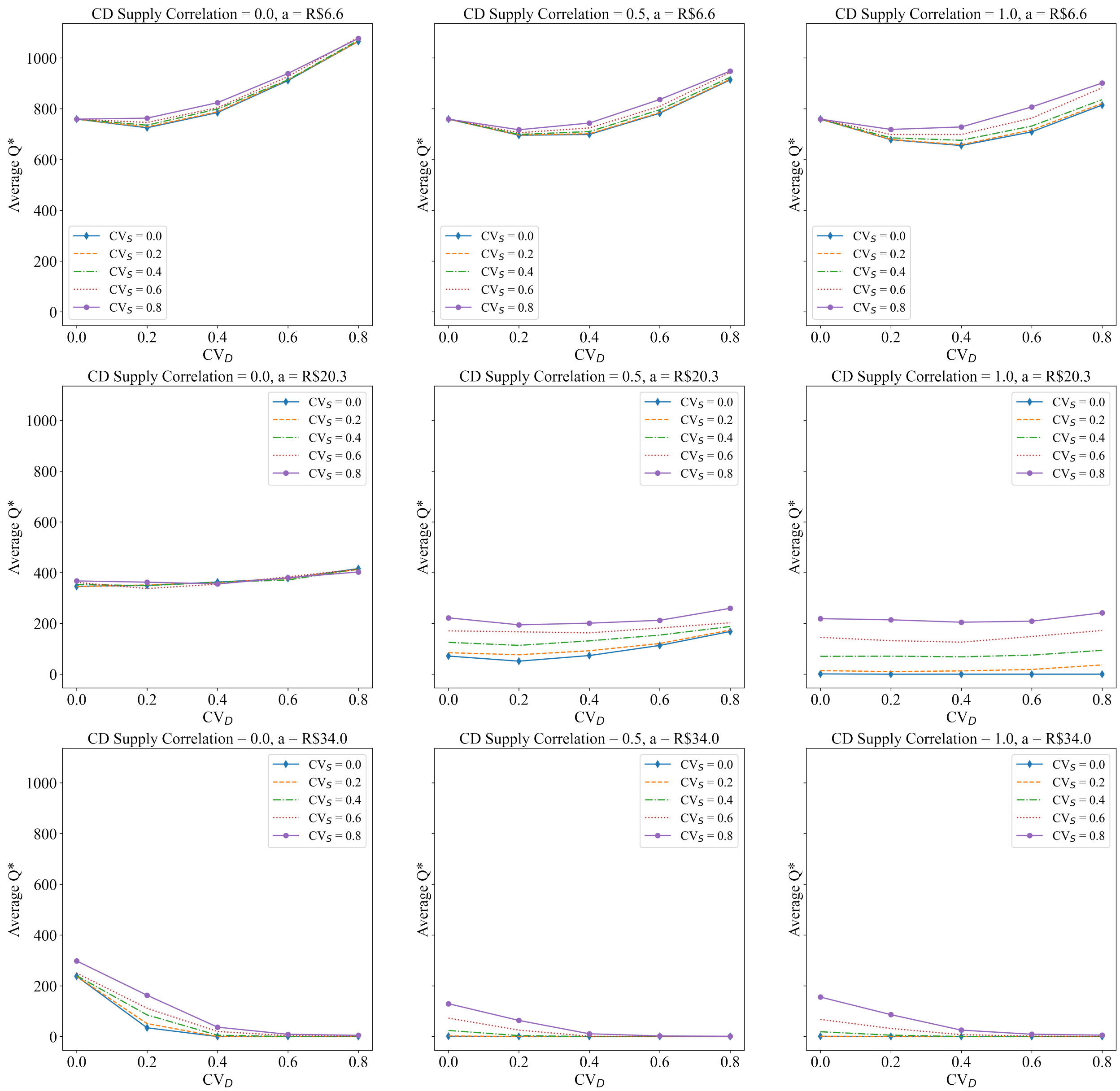

Figure S4.1: Experiment 3 Results (5x5x3x3 Factorial)

PD capacity.

Table S4.2: Full Factorial ANOVA for Experiment 3

Source of Variation	DF	F-statistic	Pr(>F)
CV_D	4	4725.0	< 0.01
CV_S	4	2876.0	< 0.01
ρ	2	38950.0	< 0.01
a	2	1005000.0	< 0.01
$CV_D \times CV_S$	16	28.2	< 0.01
$CV_D \times \rho$	8	188.8	< 0.01
$CV_S \times \rho$	8	411.0	< 0.01
$CV_D \times a$	8	5293.0	< 0.01
$CV_S \times a$	8	378.5	< 0.01
$\rho \times a$	4	7909.0	< 0.01
$CV_D \times CV_S \times \rho$	32	8.3	< 0.01
$CV_D \times CV_S \times a$	32	60.8	< 0.01
$CV_D \times \rho \times a$	16	910.1	< 0.01
$CV_S \times \rho \times a$	16	228.7	< 0.01
$CV_D \times CV_S \times \rho \times a$	64	9.6	< 0.01

Appendix S5 - Incorporating a 3PL Provider

The research question guiding this effort is concerned with finding the optimal (cost-minimizing) PD capacity when CD is available. A variation on this problem is finding the optimal privately employed capacity when also contracting for dedicated last mile delivery services through a third-party logistics (3PL) provider and CD is available. In other words, there are scenarios where a company could have three sources of LMD capacity: private capacity, dedicated capacity, and crowdsourced capacity. The adapted NVP2S model developed in this research has the flexibility to account for the additional source of LMD capacity. This appendix shows how our extended model accommodates a 3PL contracted to provide dedicated LMD services.

We assume that a firm has contracted a unit delivery rate with a Dedicated last mile 3PL for a percentage of the firm's total delivery volume. We also assume that the unit delivery cost for the dedicated 3PL, a_D, is less than the unit delivery cost for the private capacity, a: $a_D \leq a$. For this analysis, we assume the reliability of the dedicated 3PL, r_D, is slightly less than that of the private capacity, r_{Priv}, but higher than CD, r_{CD}, such that: $r_{CD} \leq r_D \leq r_{Priv}$, but the model is flexible enough to accommodate other r_D values.

The modeling process (see Table 3) can account for this scenario. Let $\tilde{D}_T$ represent the random variable of total demand and D_D represent the proportion of demand that is to be outsourced to the dedicated 3PL. The total volume of demand that is then available to optimize is $\tilde{D} = \tilde{D}_T(1 - D_D)$. The analysis can then proceed using $\tilde{D}$.

Adding a new factor into Experiment 1, which had a 5x5x3 factorial design $(CV_D \times CV_S \times \rho)$, this numerical analysis experiment considers a binary factor of including the dedicated 3PL or not, making the factorial design 5x5x3x2. The results are depicted in Figure S5.1. We expected to see Q^* generally decrease with the inclusion of a 3PL, because there is a fixed portion of LMD demand that will be allocated to the dedicated 3PL, in effect lowering the demand stochasticity that needs to be accounted for. The results confirm this expectation and demonstrate that including the 3PL lowers Q^*, the optimal PD capacity. This experiment also suggests that the impact of CV_S becomes more pronounced when CD Labor Supply Correlation is high (bottom panels) and even more so when the 3PL is included. This is likely due to the impact of removing some of the demand which will be fulfilled by the 3PL, which in turn amplifies the impact of CV_S.

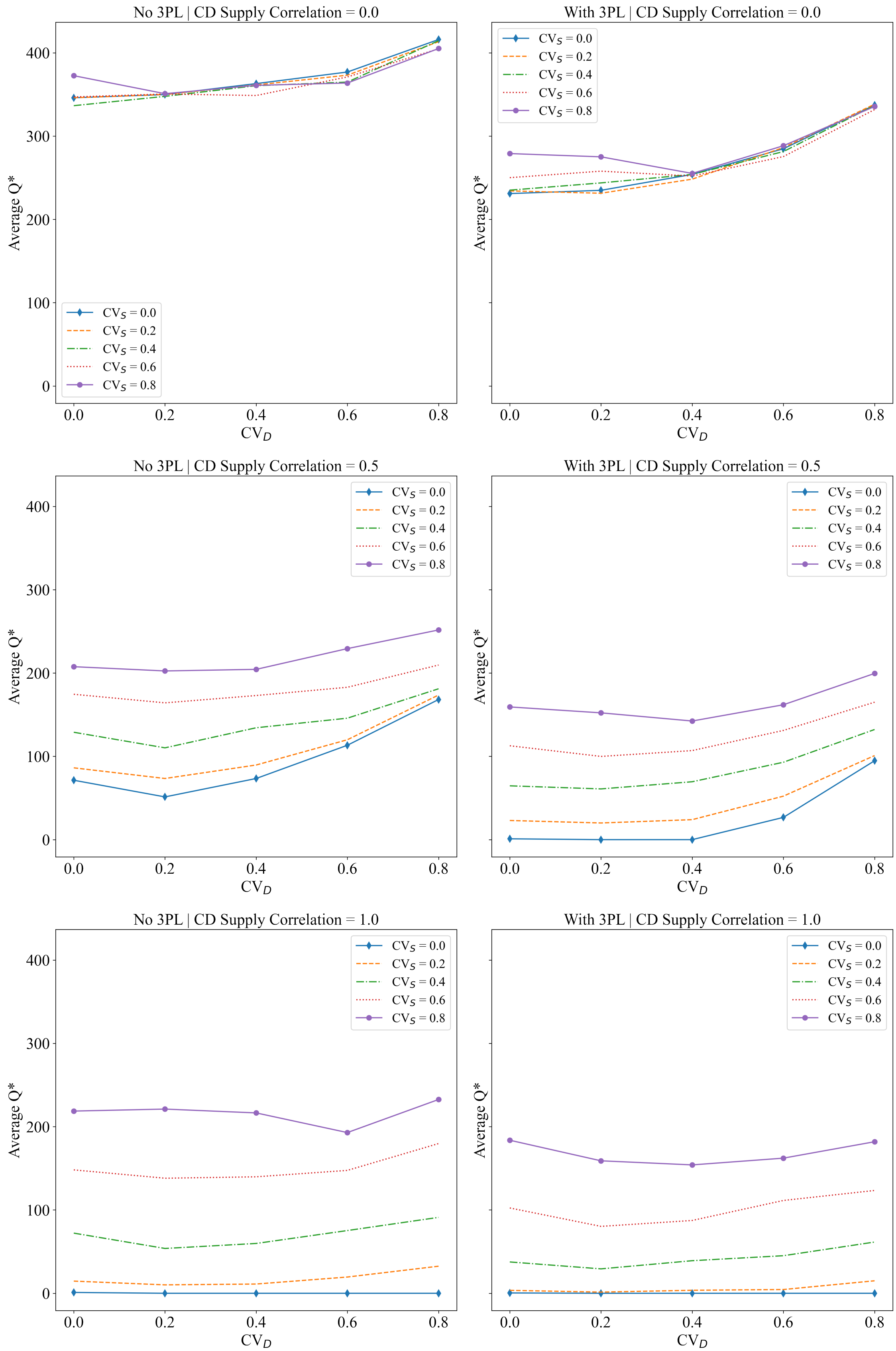

Figure S5.1: 3PL Experiment Results